5 First Order Circuits

INTRODUCTION

Transients arise in a circuit when the circuit is subjected to any change either by changing the source magnitude or by changing the circuit elements. The circuit consists of an energy storing element i.e. an inductor or capacitor.

- In the first order transients, two types of simple circuits will be examined such as a circuit consisting of resistors and capacitors only are called as RC circuit and a circuit consisting of resistors and inductors are called as RL circuit.

Gray Matter Alert!!!

First order circuits are those circuits that consists of only one energy storing element which can be either a capacitor or an inductor.

FIRST ORDER RL AND RC CIRCUITS

- In the network, if a single storage element is present then the network is described by the differential equation of first order. The RC and RL networks are considered as first order circuits.
- In this topic, following points will be taken into consideration -
  (i) Source free circuit but non-zero initial conditions.
  (ii) With source circuit.
  (iii) Circuit containing switches.

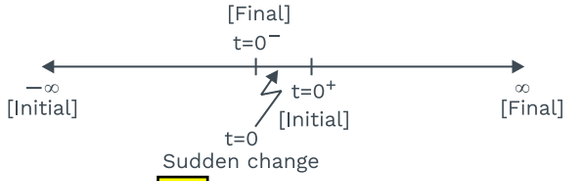


$t = 0^-$: Time instant just before the switch operation [steady state before switching]
$t = 0$: Exact time instant of switch operation
$t = 0^+$: Time instant just after switch operation
$t = \infty$: Steady state after switch operation

Definition

Transient Analysis: "The analysis of a network during an interval of time changes from one steady state condition to another is called as transient analysis".

Initial and Final Value



- Time difference between the initial and final value should always be ∞.

- At time, $t = 0^-$: Network must be drawn without sudden change.
  For example,
  At time, $t = 0$, the switch is closed
  At time, $t = 0^-$, the switch is opened
  $t = 0^-$, steady state
  → C , open circuit
  → L , short circuit
- At the time, $t = 0^+$:
  $i_L(0^+) = i_L(0^-)$
  ↳ i_L cannot be changed suddenly
  $V_C(0^+) = V_C(0^-)$
  ↳ V_C cannot be changed suddenly
  i.e. Final condition of the 1st Interval $(-\infty, 0)$ is the Initial condition of 2nd Interval $(0, \infty)$.
- For analysis at time, $t = 0^+$ replace L by current source and capacitor by voltage source.

Definition

First order circuit:
"A circuit which is characterized by the differential equation of first order is termed as first order circuits".

LAPLACE TRANSFORM

"The transform of a given derivative function in time domain (real variable 't') to a function in the frequency domain (complex variable 's') is referred to as Laplace transform".

$$\text{i.e. } L[f(t)] = F(s) = \int_{0^-}^{\infty} f(t)e^{-st}dt \quad \text{.....(1)}$$

Where, $s = \sigma + j\omega$

- Equation (1) is known as the unilateral (or one-sided) Laplace transform, the bilateral (or two-sided) Laplace transform is represented in equation (2).

$$L[f(t)] = F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt \quad \text{.....(2)}$$

- In our analysis, we will describe the circuit operation for $t \geq 0$. We do not use the bilateral from here, Thus, only one-sided Laplace transform will be used.

Rack your Brain

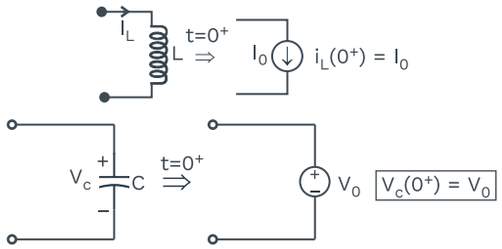
Why transients occur in electric circuits?

INITIAL AND FINAL CONDITION

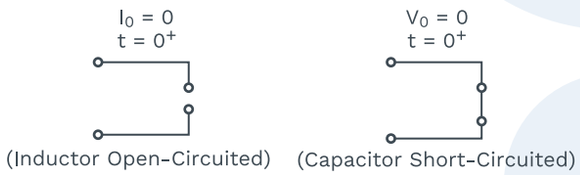
- Suppose a network exist from time $t = -\infty$ to time $t = \infty$, and a sudden change occurs in the network at time $t = 0$.

For example:

- Open or closing the switches.
- Sudden change in the value of the source then the value of a Parameter just after the sudden change [t = 0^+] is called as initial value.



- An inductor can be represented by the current source at t = 0^+ and capacitor can be represented by the voltage source but these models are only applicable at t = 0^+ and not for all values of t. If inductor and capacitor are initially uncharged.



Final Condition

- System has finally settled down and now Inductor current and capacitor voltage have become constant.

i_L = Constant

$$V_L = L\frac{di_L}{dt} = 0 \text{ [Short-circuit]}$$



V_C = Constant

$$I_C = C\frac{dV_C}{dt} = 0 \text{ [Open-circuit]}$$



Gray Matter Alert!!!

The response of a network defines the way how it reacts to an excitation.

Rack your Brain

"Inductor acts as a short circuit for DC supply." Explain?

INITIAL VALUE AND FINAL VALUE THEOREM

The initial value theorem states that-

$$f(0^+) = \lim_{t\to 0^+} f(t) = \lim_{s\to\infty} sF(s)$$

The final value theorem states that-

$$f(\infty) = \lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$$

Rack your Brain



Why does a capacitor acts as an open circuit for DC supply?

PROPERTIES OF THE LAPLACE TRANSFORM

PROPERTY	f(t)	F(s)
Linearity	$a_1f_1(t)+a_2f_2(t)$	$a_1F_1(s)+a_2F_2(s)$
Scaling	f(at)	$\frac{1}{a}F\left(\frac{s}{a}\right)$
Time Shift	f(t−a)u(t−a)	$e^{-as}\,F(s)$
Frequency Shift	$f(t)e^{s_0t}$	$F(s-s_0)$
Time Differentiation	$\frac{d^nf}{dt^n}$	$s^nF(s)-s^{n-1}f(0^-)-s^{n-2}f'(0^-)\ldots..$ $-f^{(n-1)}(0^-)$
Frequency Differentiation	tf(t)	$-\frac{d}{ds}F(s)$
Time Integration	$\int_0^t f(t)\,dt$	$\frac{1}{s}F(s)$
Frequency Integration	$\frac{f(t)}{t}$	$\int_s^\infty F(s)\,ds$
Time Periodicity	f(t)=f(t+nT)	$\frac{F(s)}{1-e^{-sT}}$
Initial Value	f(0)	$\lim_{s\to\infty} sF(s)$
Final Value	f(∞)	$\lim_{s\to 0} sF(s)$
Convolution	$f_1(t)*f_2(t)$	$F_1(s).F_2(s)$

Laplace transform of some basic signals

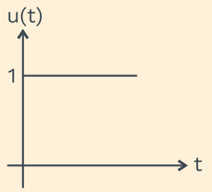
f(t)	F(s)
u(t)	$\frac{1}{s}$
$\delta(t)$	1
e^{-at}	$\frac{1}{s+a}$
t^n	$\frac{n!}{s^{n+1}}$
$t^n e^{-at}$	$\frac{n!}{(s+a)^{n+1}}$
$\sin \omega t$	$\frac{\omega}{s^2+\omega^2}$
$\cos \omega t$	$\frac{s}{s^2+\omega^2}$
$e^{-at} \sin \omega t$	$\frac{\omega}{(s+a)^2+\omega^2}$
$e^{-at} \cos \omega t$	$\frac{s+a}{(s+a)^2+\omega^2}$
$\sin(\omega t + \theta)$	$\frac{s\sin\theta + \omega\cos\theta}{s^2+\omega^2}$
$\cos(\omega t + \theta)$	$\frac{s\cos\theta - \omega\sin\theta}{s^2+\omega^2}$

Definition

Singularity Functions:
"The functions that are either discontinuous, or they have discontinuous derivatives are called as Singularity functions".

Definition

Unit Step Function:
"The unit step function, u is a function of time (t). It is '0' for the negative values of time 't' and '1' for the positive values of time 't' ".



$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$

APPLICATION OF LAPLACE TRANSFORM IN TRANSIENTS

Resistor in the s-domain

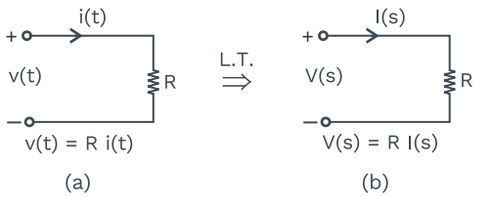


Figure : (a) Time domain representation of a resistor, (b) s-domain representation of a resistor

Inductor in the s-domain:

Consider an inductor having an initial current $i(0^+)$ in the network shown below.

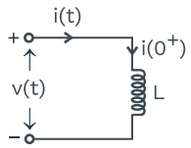


Figure 5.1: Time domain representation of charged inductor

$$v(t) = L\frac{di}{dt}$$

On taking the Laplace transform on both sides, we get-

$$V(s) = L\,[sI(s) - i(0^+)]$$

$$V(s) = sLI(s) - L.i(0^+) \qquad \text{.....(1)}$$

$$I(s) = \frac{1}{sL}V(s) + \frac{i(0^+)}{s} \qquad \text{.....(2)}$$

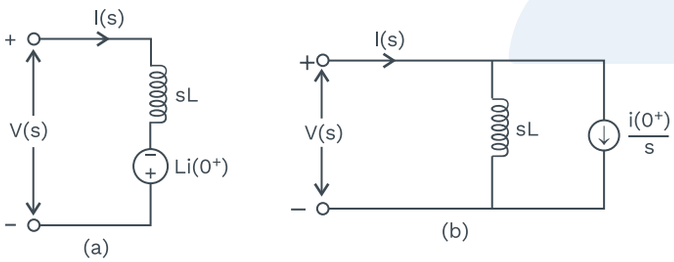


Figure 5.2: s-domain equivalent of a charged inductor

Note:

Voltage source connected in series with the inductor always supports the current.

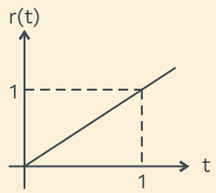
Capacitor in the s-domain:

A capacitor shown in the figure is initially charged to voltage $V(0^+)$.

Definition

Unit Ramp Function

"The unit ramp function is a function of time (t). For negative values of t, r(t) is zero and for positive values of t, r(t) has a unit slope".



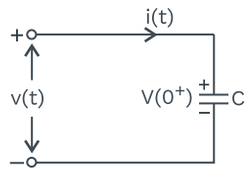


Figure 5.3: Time domain representation of a charged capacitor

$$i(t) = C\frac{dv(t)}{dt}$$

On taking the Laplace transform on both the sides, we get -

$$I(s) = C\,[s\,V(s) - V(0^+)]$$

$$I(s) = sC\,V(s) - C\,V(0^+) \qquad \text{.....(1)}$$

$$V(s) = \frac{1}{sC}\,I(s) + \frac{V(0^+)}{s} \qquad \text{.....(2)}$$

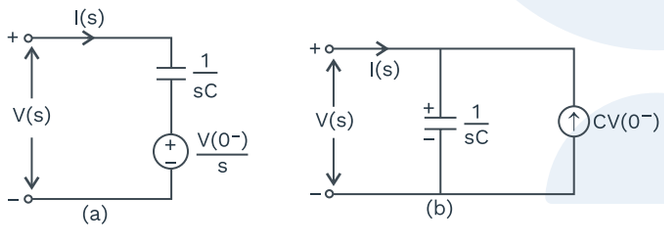


Figure 5.4: s-domain representation of a charged capacitor

Note:

In Capacitor voltage source connected in series with capacitor must be supported to capacitor voltage.

ZERO INPUT RESPONSE OR SOURCE FREE

- Source free circuits are those in which there is no source of energy and the energy is initially stored in the capacitor or inductor.

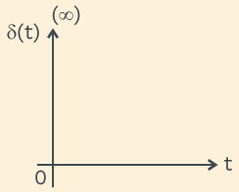
Source Free RC Circuit

"In a RC network when the DC source is suddenly disconnected, then the circuit is known as source free RC circuit". So, the energy which is already stored in a capacitor is thus transferred to the resistor present in the given circuit-

Definition

Unit Impulse Function:

"The unit impulse function is a function of time (t). δ(t) is zero everywhere except for time t = 0, where it is undefined".



Rack your Brain

Why does not a capacitor allow sudden changes in the voltage?

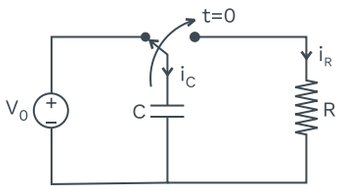
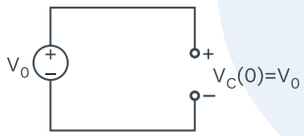


Figure 5.5: Source free RC circuit

The capacitor voltage in the above circuit is $v_C(t)$. At time t = 0, the capacitor is initially charged, the capacitor voltage is-

$$v_C(0) = V_0 \qquad \text{.....(1)}$$

At time, $t = 0^-$, the circuit is -

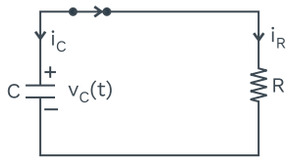


Initial condition: $V_C(0^-) = V_C(0^+) = V_C(0) = V_0$

The energy stored is-

$$W_C(0) = \frac{1}{2} C V_0^2 \qquad \text{.....(2)}$$

For time, t > 0



Now applying KCL in the above circuit-

$$i_C + i_R = 0$$

By definition-

$$i_C = C\frac{dv_C}{dt}, \text{ and } i_R = \frac{v_C}{R}.$$

Thus,

$$C\frac{dv_C}{dt} + \frac{v_C}{R} = 0 \text{ or } \frac{dv_C}{dt} + \frac{v_c}{RC} = 0$$

Rack your Brain

Why does a capacitor oppose instantaneous change in the voltage?

The above equation represents the first derivative of the capacitor voltage (v_C). So, the equation is a differential equation of first order.
The equation can be rearranged as-

$$\frac{dv_C}{v_C} = -\frac{1}{RC}dt$$

On integrating both the sides, we get-

$$\int_{V_0}^{v_C(t)} \frac{dv_C}{v_C} = -\int_0^t \frac{1}{RC}dt$$

$$\ln\left(v_C\right)\Big|_{v_0}^{v_C(t)} = -\frac{t}{RC}\Big|_0^t$$

$$\ln\frac{v_C(t)}{V_0} = -\frac{t}{RC}$$

$$v_C(t) = V_0 e^{-\frac{t}{RC}} \qquad \text{.....(3)}$$

$$i_R(t) = \frac{v_C(t)}{R} = \frac{V_0}{R}e^{-\frac{t}{RC}} \qquad \text{.....(4)}$$

- Equation (3) shows the voltage response of a RC circuit, which represent the exponential decay of an initial voltage.
- The response of the circuit is obtained due to an initial energy stored in the circuit which is called natural response of a the circuit.
- The figure 5.6 shows the natural response of a circuit. For time t = 0, the initial conditions are represented. As the time increases, the voltage decreases to zero. The rate of decrease in voltage, is expressed by the time constant, denoted by τ.
- "Time required for the response in a circuit to decay to a factor of 1/e or by 36.8 percent of the initial value, is referred to as time constant".

Rack your Brain

Why it is possible to change the instantaneous voltage across a resistor?

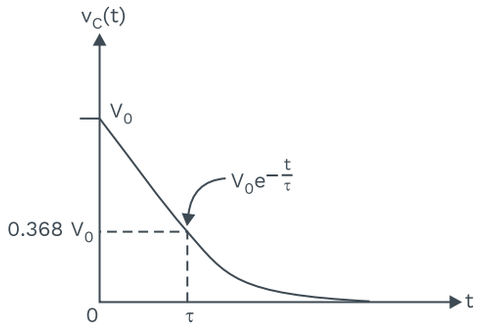


Figure 5.6: Voltage response of a RC circuit.

In terms of the time constant τ, the equation (3) and (4) are-

$$v_C(t) = V_0 e^{-\frac{t}{\tau}} \quad \text{.....(5)}$$

$$i_R(t) = \frac{V_0}{R} e^{-\frac{t}{\tau}} \quad \text{.....(6)}$$

where τ = time constant

$\tau = RC$

t	$v_C(t)$	$v_C(t)/V_0$
τ	0.3678 V_0	0.3678
2τ	0.1353 V_0	0.1353
3τ	0.0497 V_0	0.0497
4τ	0.0183 V_0	0.0183
5τ	0.0067 V_0	0.0067

Note: For every interval of its previous value $v_C(t + \tau) = \frac{v_C(t)}{e} = 0.3687\, v_C(t)$, regardless of the value of t.

At $t = 3\tau$, $v_C(t) < 5\%\ V_0$.

At $t = 5\tau$, $v_C(t) < 1\%\ V_0$.

In first order circuits, settling time = 5τ

Gray Matter Alert!!!

The time constant is also known as the time delay. Time required for a capacitor in a RC circuit to be fully charged is equal to 5 time constant.

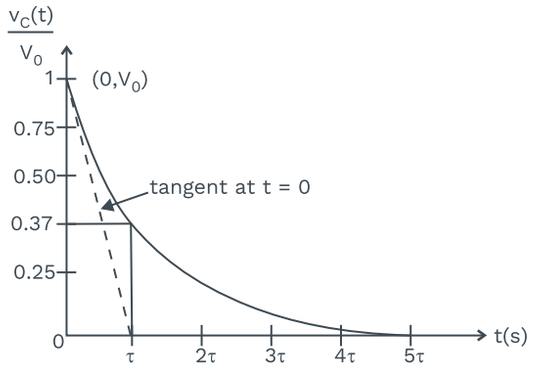


Figure 5.7: Graphical representation of the time constant from the response curve

Graphical Interpretation:

$$\frac{dv_C(t)}{dt} = \frac{d}{dt} V_0 e^{-\frac{t}{\tau}}$$

$$= \frac{-V_0}{\tau} e^{-\frac{t}{\tau}}$$

Slope of tangent = slope of $v_C(t)$ at t = 0

$$= -\frac{V_0}{\tau}$$

$$(Y - V_0) = \frac{-V_0}{\tau} (t - 0)$$

$$Y = v_C(t) = V_0 - V_0 \frac{t}{\tau}$$

x-intercept of tangent,

$v_C(t) = 0$

$V_0 = V_0 \frac{t}{\tau}$

$t = \tau$

- If the tangent is drawn to the capacitor voltage at t = 0, then it intersects the steady state value [$v_C(t) = 0$] at $t = \tau$.

Note:
The rate of decay in a circuit depends on the time constant (τ), the faster the rate of decay, the lesser will be the time constant τ and vice-versa.

Rack your Brain

What are the applications of an RC circuit?

$$i_C(t) = C\frac{dv_C(t)}{dt}$$

$$= C\frac{d}{dt}\left(V_0 e^{-\frac{t}{\tau}}\right)$$

$$i_C(t) = -\frac{C}{\tau}V_0 e^{-\frac{t}{\tau}} \qquad [\tau = RC]$$

$$i_C(t) = -\frac{V_0}{R}e^{-\frac{t}{\tau}} \qquad \text{.....(7)}$$

Note :

The polarity of the voltage across the capacitor remains the same but the current direction gets reversed when the capacitor is in discharged condition.

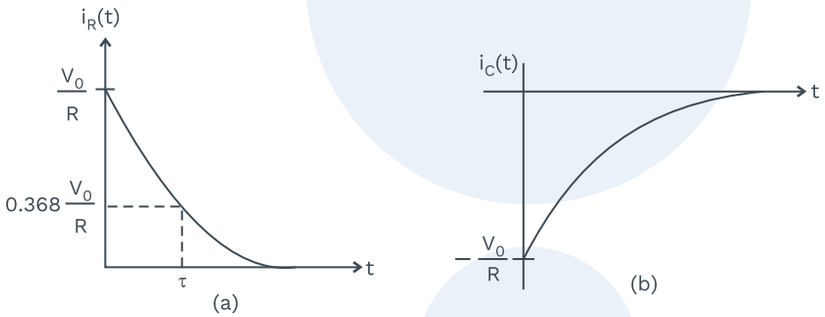


Figure 5.8 : Current response of a RC circuit

Energy Consideration

The dissipated power in the resistor R is,

$$P(t) = i_R^2(t)R = \frac{V_0^2}{R}e^{-\frac{2t}{\tau}} \qquad \text{.....(8)}$$

At any time t, the energy stored in the capacitor is given by-

$$W_C(t) = \frac{1}{2}C v_C^2(t) = \frac{1}{2}C\left[V_0 e^{-\frac{t}{\tau}}\right]^2$$

$$W_C(t) = \frac{1}{2}CV_0^2 e^{-\frac{2t}{\tau}} \qquad \text{.....(9)}$$

- By Energy Conservation-

Energy dissipated in R = Energy lost by C

$$W_R(t) = W_C(0) - W_C(t)$$

$$= \frac{1}{2}C V_0^2 - \frac{1}{2}CV_0^2 e^{-\frac{2t}{\tau}}$$

$$W_R(t) = \frac{1}{2} C V_0^2 \left[1 - e^{-\frac{2t}{\tau}}\right] \quad \text{.....(10)}$$

When $t \to \infty$, $W_R(\infty) \to \frac{1}{2} C V_0^2$,

The above equation represents the energy stored in a capacitor initially is dissipated to the resistor.

Steps required for working with a source-free RC circuit are-

Step 1: Find the initial voltage across the capacitor before t = 0. Since the DC source is connected for a long time so before t = 0 the circuit would be at its steady state.
To obtain the capacitor's voltage, assume the circuit for time t < 0, the capacitor behaves as an open circuit and then finds the voltage across the terminals of a capacitor.

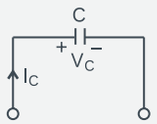
Step 2: Now assume the circuit for time t > 0, to obtain the time constant τ = RC, resistance R is taken as the Thevenin equivalent resistance across the terminals of a capacitor.
So, the time constant will be $\tau = R_{Th}$.

Time Constant:

"In a circuit the time required for the response to decay to a factor of $\frac{1}{e}$ or 36.8% of the initial value while discharging or the time taken by the response to reach 63.2% of its final value while charging is referred to as time constant, and it is denoted by τ".

Gray Matter Alert!!!

Consider the given case of capacitor.



(a) If voltage V_C decreases with time, I_C becomes negative. So, in this case, power is delivered by the capacitor.

(b) If voltage V_C increases with time, I_C become positive. So, in this case, power is absorbed by a capacitor.

Rack your Brain

What is the importance of time constant in an RC circuit?

Total response = Natural Response + Forced Response

Natural Response → ZIR → For transient → Complementary function (CF)

Forced Response → ZSR → For steady state → Particular integral (PI)

Where,

ZIR = zero-input response

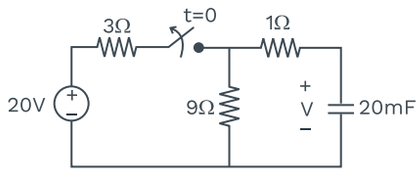
ZSR = zero-state response

An electrical network has two type of responses such as-

- "The response of a network without any source is called a natural response" and this response leads to a transient response.
- The response of a network with a source present in it is called as forced response and it leads to the steady state response, this response depends on the type of an input which is obtained by solving the particular integral (PI) part of differential equation.

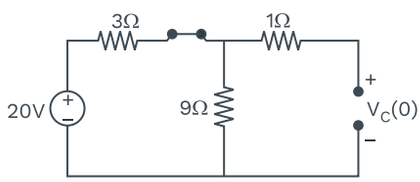
Example 1:

For the circuit shown below, the switch is closed for a very long time and then it is open for time t = 0. Obtain the voltage v(t) for time t ≥ 0 and the initial energy stored in the capacitor.



Solution:

For time t < 0, the switch is closed and the capacitor behaves as an open circuited, as shown in figure (a).



(a)

Using the voltage division rule, we get-

$$V_C(0) = 20 \times \frac{9}{3+9} = 15V, t < 0$$

Since the voltage in a capacitor cannot be changed suddenly, so the capacitor's voltage for time t = 0^- is same as for time t = 0.

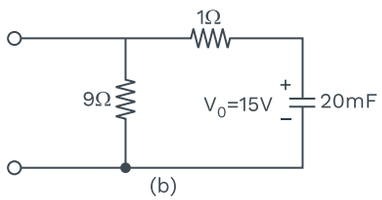
i.e. $V_C(0) = V_0 = 15$ V

The switch is open for time t > 0, the RC network is represented in figure (b).

In the figure (b) the circuit is source free, so an independent source is used to give the initial energy to the capacitor.

The 1 Ω resistor and 9 Ω resistor are in series, so the equivalent resistance is-

$R_{eq} = 1 + 9 = 10\ \Omega$



(b)

The time constant of the network is-

$\tau = R_{eq} C = 10 \times 20 \times 10^{-3}$

$\tau = 0.2$ seconds

The voltage across the capacitor for time t ≥ 0 is-

$$v_C(t) = V_C(0) e^{-\frac{t}{\tau}}$$

$$= 15e^{-\frac{t}{0.2}} V$$

or $v_C(t) = 15e^{-5t}$ V

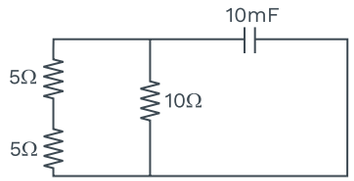
The initial energy stored in the capacitor is-

$$W_C(0) = \frac{1}{2} C V_C^2(0) = \frac{1}{2} \times 20 \times 10^{-3} (15)^2 = 2.25 \text{ J}$$

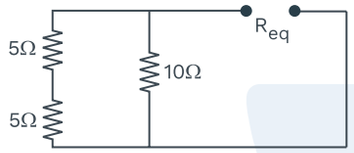
Rack your Brain

What is the significance of a switch in the transient analysis of electric circuits?

Example 2:
In the given network, find the time constant (τ).



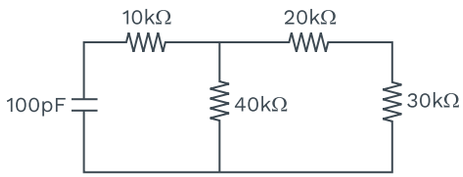
Solution:
For R_{eq}:



R_{eq} = (5+5) || 10
= 5 Ω
C = 10 mF

Time constant for RC circuit is-
$\tau = R_{eq}\, C$
$= 5 \times 10 \times 10^{-3}$
= 50 ms

Example 3:
Obtained the time constant (τ) for the circuit is shown below.



Solution:
Time constant, $\tau = R_{Th}C$
Where,
R_{Th} = Thevenin equivalent at the capacitor terminals

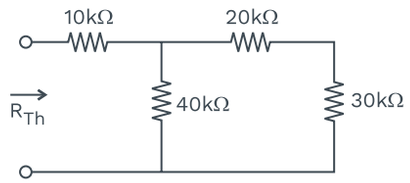
Previous Year's Questions

If the Laplace transform of the voltage across a capacitor of value of $\frac{1}{2}$F is-

$$V_C(s) = \frac{s+1}{s^3+s^2+s+1}$$

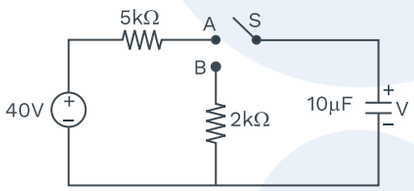
The value of the current through capacitor at $t = 0^+$ is-

(1) 0 A (2) 2 A
(3) $\frac{1}{2}$A (4) 1 A

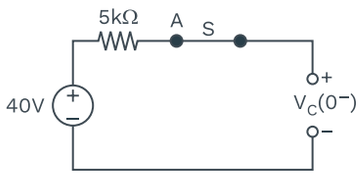


R_{Th} = 10 + (20 + 30) || 40
= 32.22 kΩ
$\tau = R_{Th} C$
$\tau = 32.22 \times 10^3 \times 100 \times 10^{-12}$
τ = 3.22 μs

Example 4:
For the circuit shown below, the switch S is in position A for a very long time at time t = 0, the switch S moves from A position to B position. Calculate the voltage across the capacitor (V_C) for time t > 0.



Solution:
For time t < 0,



$V_C(0^-) = 40$ V

The voltage across the capacitor cannot be changed suddenly.
$\therefore \quad V_C(0^-) = V_C(0) = V_C(0^+) = 40$ V
For time t > 0, we have a source-free RC circuit.



The time constant is-

$$\tau = R_{Th} C = 2 \times 10^3 \times 10 \times 10^{-6}$$

or $\tau = 0.02$ seconds

Thus, the voltage (V) across the capacitor for time $t > 0$ is-

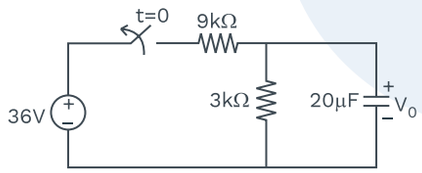
$$V(t) = V_C(0^+)e^{-\frac{t}{\tau}}$$

$$V(t) = 40e^{-\frac{t}{0.02}}$$

or $V(t) = 40e^{-50t}\,V$

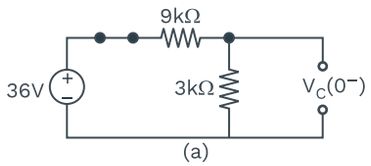
Example 5:

Determine the voltage $V_0(t)$ for time $t > 0$ as shown in the figure and also find the time required for a capacitor voltage to decay to one-third of its value for time $t = 0$.



Solution:

For time $t < 0$, the switch in the circuit is closed, the capacitor acts as an open circuit, as represented in the figure (a).

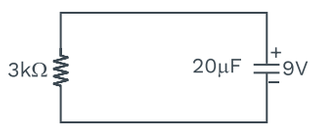


(a)

$$V_C(0^-) = (36)\frac{3}{3+9} = 9\,V$$

$\therefore V_C(0^-) = V_0 = 9V$

For $t > 0$, we have a source-free RC circuit, as represented in the figure (b).



(b)

The time constant (τ) is-

$$\tau = RC = 3 \times 10^3 \times 20 \times 10^{-6}$$
$$= 60 \times 10^{-3}$$
$$= 0.06 \text{ seconds}$$

$$V_0(t) = V(0)e^{-\frac{t}{\tau}}$$

$$= 9e^{-\frac{t}{0.06}}$$

$$V_0(t) = 9e^{-16.67t} \text{ V} \qquad \text{.....(1)}$$

Let the time be t_0,

$$3 = 9e^{-16.67 t_0}$$

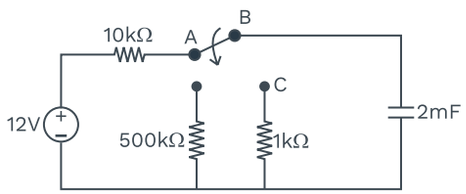
or $$e^{-16.67 t_0} = \frac{1}{3}$$

or $16.67\, t_0 = \ln 3$

$$t_0 = \frac{\ln 3}{16.67} = 65.9 \text{ milliseconds}$$

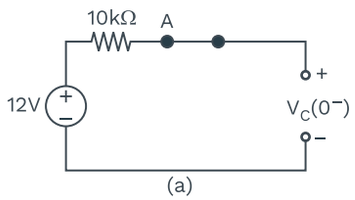
Example 6:

The switch shown in the figure is in position 'A' for a very long time and is moved to position 'B' at time t = 0, then at time t = 1 second, the switch moves from position 'B' to position 'C'. Determine the voltage across the capacitor for time $t \geq 0$.



Solution:

For time t < 0, the switch is at position (A) by the circuit reaches to the steady state, as represented in the figure (a).

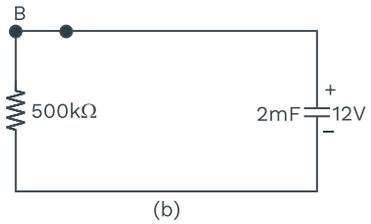


(a)

$V_C(0^-) = 12V$

$V_C(0^-) = V_C(0) = V_C(0^+) = 12V$

For time $0 < t < 1$, the switch is at position B, as represented in the figure (b).



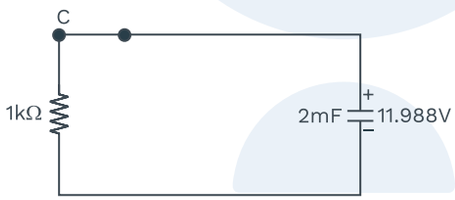
(b)

Time constant (τ) = RC

$= 500 \times 10^3 \times 2 \times 10^{-3} = 1000$ seconds

$V_C(t) = 12e^{-\frac{t}{1000}} = 12e^{-0.001t}$ volts

For $1 < t < \infty$ the switch is at position (C) as represented in the figure (c).



$V_C(1) = 12e^{-\frac{1}{1000}} = 11.988V$

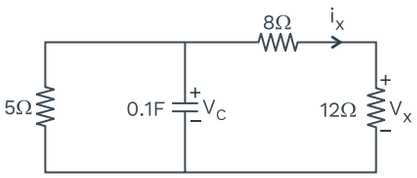
$V_C(1) = V_C(1^-) = V_C(1^+) = 11.988$ V

$t = RC = 1 \times 10^3 \times 2 \times 10^{-3} = 2$ seconds

$V_C(t) = 11.988e^{-\frac{(t-1)}{2}} = 11.988e^{-0.5(t-1)}$ volts

Example 7:

For the circuit shown below, the initial voltage across the capacitor is $V_C(0) = 15V$. Obtain the voltages V_C, V_x and current i_x for time $t > 0$.



Solution:

In the given circuit, the two resistors 8Ω and 12 Ω are in series, so their combination gives 20 Ω resistor. Then, the 20 Ω resistor will be in parallel with the 5 Ω resistor. So the equivalent resistance in the circuit is-

$$R_{eq} = \frac{20 \times 5}{20 + 5} = 4\Omega$$

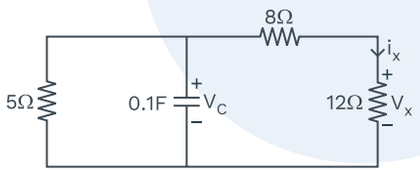
The time constant (τ) is-

$$\tau = R_{eq}\, C = 4 \times 0.1 = 0.4 \text{ seconds}$$

Thus,

$$V = V(0)\, e^{-\frac{t}{\tau}} = 15e^{-\frac{t}{0.4}}$$

$$V_C = V = 15e^{-2.5t} \text{ Volts}$$



On applying voltage division rule, V_x is-

$$V_x = V\left(\frac{12}{12 + 8}\right)$$

$$= 0.6 \text{ V}$$

$$= 0.6\ (15e^{-2.5t})$$

$$= 9e^{-2.5t} \text{ Volts}$$

$$\text{and } i_x = \frac{V_x}{12} = \frac{9}{12} e^{-2.5t}$$

$$i_x = 0.75e^{-2.5t} \text{ A}$$

Source-free RL Circuit

"When a DC source in a circuit is suddenly disconnected, then the circuit is referred as a source-free RL circuit". So the energy in an inductor is transferred to the resistor present in the circuit as shown below.

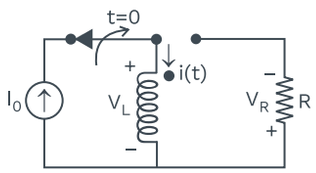
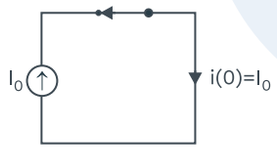


Figure 5.9: Source-free RL circuit

- Consider the circuit shown in the figure (5.9), before time t = 0, the inductor was connected to an ideal current source and an inductor gets charged by a current I_0 and at time t = 0, the switch is moved to make connection between the resistor and a charged inductor. To determine the circuit response, assume a current i(t) through the inductor.
  At time t = 0, consider the initial current in an inductor is-

  $$i(0) = I_0 \qquad \text{.....(1)}$$

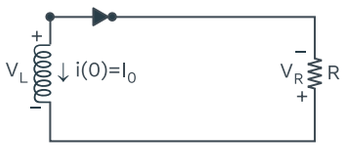
- At time $t = 0^-$, the switch in the circuit is closed, the inductor acts as a short circuit to DC, as shown in the figure below.



  Initial condition, $i(0^-) = i(0) = i(0^+) = I_0$ with the corresponding energy stored in the inductor as –

  $$W_L(0) = \frac{1}{2}LI_0^2 \qquad \text{.....(2)}$$

- For t = 0, the switch is moved to make the connections between resistor and inductor.
  ∴ For t > 0,



  Now applying KVL for t > 0 in the circuit-

  $$V_L + V_R = 0$$

  $$L\frac{di}{dt} + iR = 0 \text{ or } \frac{di}{dt} + \frac{R}{L}i = 0$$

  The above equation is a first-order differential equation. As only the first derivative of current 'i' is involved.

Rack your Brain

Why does an inductor oppose instantaneous change in the current?

The equation can be rearranged as-

$$\frac{di}{dt} = -\frac{R}{L}i \text{ or } \frac{di}{i} = -\frac{R}{L}dt$$

On integrating both the sides, we get-

$$\int_{I_0}^{i(t)} \frac{di}{i} = -\int_0^t \frac{R}{L}dt$$

$$\ln i\Big|_{I_0}^{i(t)} = -\frac{Rt}{L}\Big|_0^t$$

$$\ln\left[\frac{i(t)}{I_0}\right] = -\frac{R}{L}t$$

$$i(t) = I_0 e^{-\frac{Rt}{L}} \quad \text{.....(3)}$$

Equation (3) represents that the natural response of a RL circuit is an exponential decay of its initial current.
Then the time constant for a RL circuit is-

$$\tau = \frac{L}{R} \quad \text{.....(4)}$$

Thus, equation (3) may be written as:

$$i(t) = I_0 e^{-\frac{t}{\tau}} \quad \text{.....(5)}$$

At $t = \tau$, $i = 0.368\ I_0$

So, the inductor current becomes 36.8% value of the initial value.

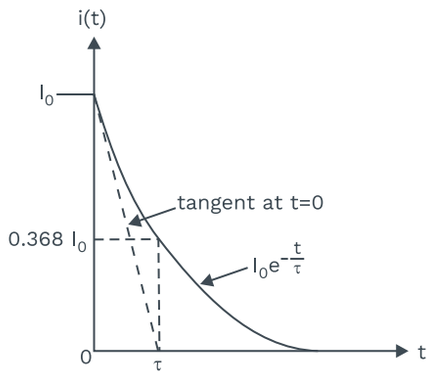


Figure 5.10: Current response of a RL circuit

Using equation (5), the voltage across the resistor and an inductor can be determined as-

$$V_R(t) = i(t)R$$

$$V_R(t) = I_0 R e^{-\frac{t}{\tau}} \quad \text{.....(6)}$$

$$V(t) = L\frac{di(t)}{dt}$$

$$= L\frac{d}{dt}\left(I_0 e^{-\frac{t}{\tau}}\right)$$

$$= -\frac{I_0 L}{\tau} e^{-\frac{t}{\tau}}$$

$$V(t) = -I_0 R e^{-\frac{t}{\tau}} \quad \text{.....(7)}$$

Note:

discharging an inductor the current direction does not change, but polarity of the voltage across an inductor gets reversed.

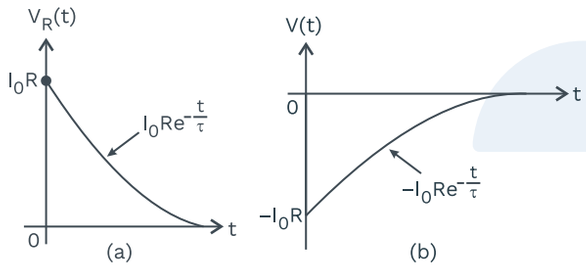


Figure 5.11: Voltage response of RL circuit

Rack your Brain

How much time is required for a current in an RL circuit to decay to zero?

Energy Considerations

- The power dissipated in the resistor R is-

$$P(t) = i^2(t)\ R$$

$$= I_0^2 e^{-\frac{2t}{\tau}} . R$$

$$P(t) = I_0^2 R e^{-\frac{2t}{\tau}} \quad \text{.....(8)}$$

- At any time t energy stored in the inductor by-

$$W_L(t) = \frac{1}{2} L i^2(t)$$

$$= \frac{1}{2}L\left(I_0 e^{-\frac{t}{\tau}}\right)^2$$

$$= \frac{1}{2}LI_0^2 e^{-\frac{2t}{\tau}}$$

or $W_L(t) = \frac{1}{2}LI_0^2 e^{-\frac{t}{0.5\tau}}$ .....(9)

Where, $\tau_{energy} = \frac{\tau}{2}$

So, energy decays faster than current.

- By energy conservation,

Energy dissipated in R = Energy lost by L

$$W_R(t) = W_L(0) - W_L(t)$$

$$= \frac{1}{2}LI_0^2 - \frac{1}{2}LI_0^2 e^{-\frac{2t}{\tau}}$$

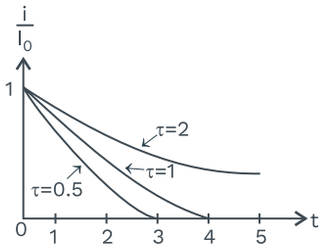
$$W_R(t) = \frac{1}{2}LI_0^2\left[1 - e^{-\frac{2t}{\tau}}\right] \quad .....(10)$$

As $t \to \infty$, $W_R(\infty) \to \frac{1}{2}LI_0^2$,

The above equation is the same as $W_L(0)$, thus the power dissipated in a resistor is the initial energy that was stored in the inductor.

Note:

The rate of decay of the response depends on the time constant (τ) of a circuit. The faster the rate of decay, the smaller the time constant (τ) of the circuit is-



Rack your Brain

What are the applications of an RL circuit?

Steps required for a source-free RL circuit are-

Step 1: Find the initial current stored in the inductor for t < 0. Since the DC source is connected for long time so before t = 0 the circuit would be at its steady state. In steady state inductor will be short circuit and then obtain the current at the terminals of an inductor.

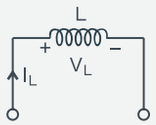
Step 2: Now consider circuit t > 0 (after t = 0) and finding the time constant $\tau = \frac{L}{R}$, R is the Thevenin equivalent resistance at the terminals of an inductor, R = R_{Th} at the terminals.

So, the time constant will be:

$$\tau = \frac{L}{R_{Th}}$$

Gray Matter Alert!!!

Consider the given case of an inductor-

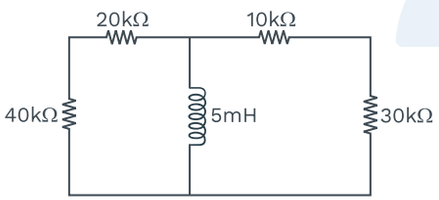


(a) If current I_L decrease with time, V_L become negative. So, in this case, power is delivered by the inductor.

(b) In current I_L increase with time, V_L become positive. So, in this case, power is absorbed by the inductor.

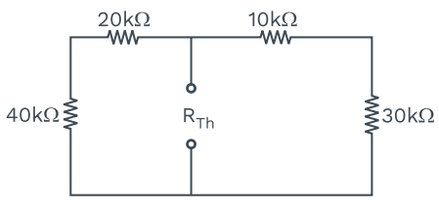
Example 8:

For the given network, find the time constant (τ).



Solution:

For R_{Th} :



Thevenin resistance seen at terminal of inductor.

R_{Th} = (20+40) || (10+30)

= (60 || 40)

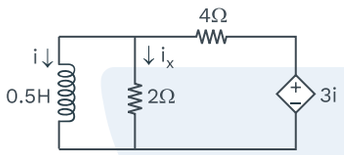
$$= \frac{60 \times 40}{60 + 40} = 24k\Omega$$

Time constant, $\tau = \frac{L}{R_{Th}}$

$$= \frac{5 \times 10^{-3}}{24 \times 10^{3}} = 0.208\mu s$$

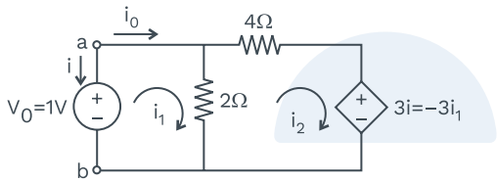
Example 9:

For the circuit shown in the figure, assume that the initial value of current is i(0) = 10 A. Determine the value of current i(t) and i_x(t).



Solution:

Due to the dependent source, a voltage source of V_0 = 1V is attached at the terminals a and b as shown in the figure below.



Applying KVL in the above circuit, we get-

$$-1 + 2(i_1 - i_2) = 0$$

$$2i_1 - 2i_2 = 1$$

$$i_1 - i_2 = 0.5 \quad \text{.....(1)}$$

$$4i_2 - 3i_1 + 2(i_2 - i_1) = 0$$

$$-5i_1 + 6i_2 = 0$$

$$i_2 = \frac{5i_1}{6} \quad \text{.....(2)}$$

Putting the value of i_2 in the equation (1), we get-

$$i_1 - \frac{5i_1}{6} = 0.5$$

$$i_1 = 3A$$

$$i_0 = i_1 = 3A$$

$$R_{Th} = \frac{v_0}{i_0} = \frac{1}{3}\Omega$$

The time constant is-

$$\tau = \frac{L}{R_{eq}} = \frac{L}{R_{Th}} = \frac{0.5}{1/3} = \frac{3}{2}\text{seconds}$$

Thus, current through inductor is-

$$i(t) = i(0)e^{-\frac{t}{\tau}}$$

$$= 10e^{-(2/3)t}\ A, \text{ for } t > 0$$

The voltage across an inductor is-

$$V = L\frac{di}{dt} = 0.5(10)\left(-\frac{2}{3}\right)e^{-\left(\frac{2}{3}\right)t}$$

$$= -\frac{10}{3}e^{-\left(\frac{2}{3}\right)t}V$$

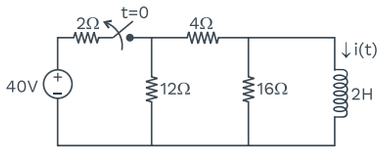
Since the inductor and the 2Ω resistor present in the circuit are in parallel,

$$i_x(t) = \frac{V}{2} = -\frac{10}{2\times 3}e^{-\left(\frac{2}{3}\right)t}$$

or $i_x(t) = -1.6667\ e^{-(2/3)t}\ A,\ t > 0$

Example 10:

In the network, the switch is closed for time t < 0. At time t = 0 the switch is opened. Obtain the value of current i(t) for time t > 0.



Solution:

For time t < 0, the inductor behaves as a short circuit. Thus the 16Ω resistor is ignored due to short-circuited of inductor. To obtain current i_1 in the figure (a), 4Ω and 12Ω resistor are in parallel to each other. So,

Rack your Brain

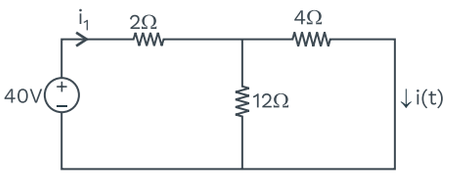
In a series RL circuit, when the switch is closed initially and the circuit is complete then the inductor in the circuit behave as-

(A) Capacitor (B) Resistor

(C) Short-circuit (D) Open-circuit

$$\frac{4 \times 12}{4 + 12} = 3\Omega$$

Hence, $i_1 = \frac{40}{2 + 3} = 8A$



(a)

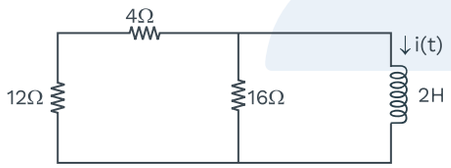
By using current division rule, we get-

$$i(t) = 8\left(\frac{12}{12 + 4}\right) = 6A,\ t < 0$$

Since the current in an inductor is-

$$i(0) = i(0^-) = i(0^+) = 6A$$

For time t > 0, the switch is open and thus the voltage source will be disconnected.
So the source-free RL circuit is shown as-



(b)

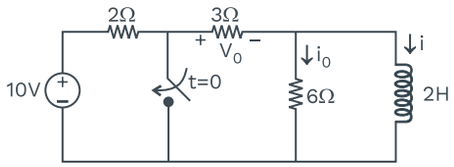
For equivalent resistance of the circuit -
$R_{eq} = (12 + 4) \parallel 16$
$R_{eq} = 8\ \Omega$

The time constant for RL circuit is-

$$\tau = \frac{L}{R_{eq}} = \frac{2}{8} = \frac{1}{4} \text{ seconds}$$

Thus, $i(t) = i(0)e^{-t/\tau}$
$i(t) = 6e^{-4t}\ A$

Example 11:
For the circuit shown in the figure, obtain the value of current i_0 and i for all the time, assume the switch in the circuit is opened for a very long time. Also determine the voltage v_0.

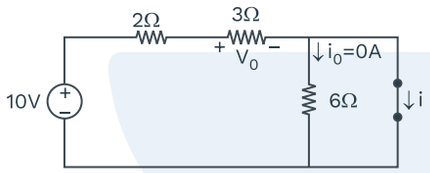


(b)

Solution:

For time t < 0, the switch is open, since the inductor acts as a short circuit to DC then the 6Ω resistor will be ignored due to short-circuited of inductor as shown figure (a).

So, i_0 = 0A.



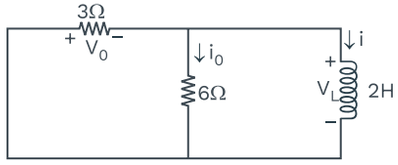
(a)

$$i(t) = \frac{10}{2+3} = 2A,\ t < 0$$

$V_0(t) = 3i\ (t) = 3\times2 = 6V,\ t < 0$

Thus, the initial value of current i(0) = 2A

For time t > 0, the switch in the circuit is closed, so that the voltage source will be short-circuited. Thus, the circuit is a source-free RL circuit as shown below.



(b)

At the inductor terminals,

$R_{Th} = (3 \parallel 6) = 2\Omega$

The time constant (τ) is-

$$\tau = \frac{L}{R_{Th}} = 1 \text{ seconds}$$

Hence, $i(t) = i(0)\ e^{-t/\tau}$

$i(t) = 2e^{-t}$ A, t > 0

Since, the 6Ω and 3Ω resistors are in parallel with an inductor,

$$V_0(t) = -V_L = -L\frac{di}{dt}$$

$$V_0(t) = -2 \times [2 \times (-1) \times e^{-t}]$$

$$V_0(t) = 4e^{-t}\ V,\ t > 0$$

$$\text{and } i_0(t) = \frac{V_L}{6} = \frac{-4e^{-t}}{6} = \frac{-2}{3}e^{-t}A$$

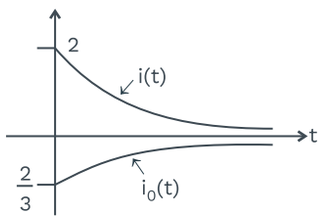
$$i_0(t) = \frac{-2}{3}e^{-t}A,\ t > 0$$

Thus, for all time,

$$i_0(t) = \begin{cases} 0A, & t < 0 \\ \frac{-2}{3}e^{-t}A, & t > 0 \end{cases}$$

$$V_0(t) = \begin{cases} 6V, & t < 0 \\ 4e^{-t}V, & t > 0 \end{cases}$$

$$i(t) = \begin{cases} 2A, & t < 0 \\ 2e^{-t}, & t > 0 \end{cases}$$

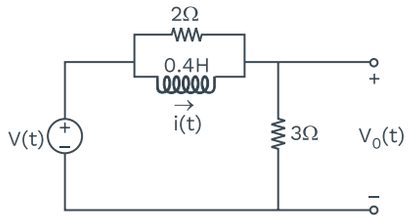


The current through an inductor is continuous for time t = 0, whereas the current through a 6Ω resistor drops from 0A to $\frac{-2}{3}$A at time t = 0, and the voltage drops across the 3Ω resistor from 6V to 4V at time t = 0.
From the above statements the time constant is same regardless of the output.

Example 12:
Consider the circuit in the figure, determine $V_0(t)$, when i(0) = 5A and V(t) = 0.

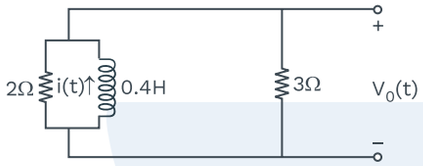
Gray Matter Alert!!!

In a steady state circuit, the current and the voltage do not change with time and the amplitude or frequency of a sinusoid never changes.



Solution:

If V(t) = 0, the circuit can be redrawn as shown below.



The equivalent resistance R_{eq} is-

$$R_{eq} = R_{Th} = (2 \parallel 3) = \frac{2 \times 3}{2 + 3} = \frac{6}{5}\Omega$$

The time constant (τ) is-

$$\tau = \frac{L}{R_{Th}} = \frac{0.4}{6/5} = \frac{1}{3}\text{seconds}$$

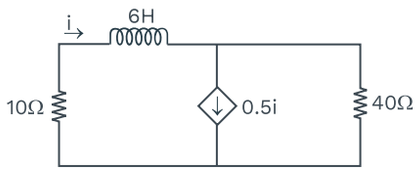
$$i(t) = i(0)e^{-t/\tau} = 5e^{-3t}$$

$$V_0(t) = -L\frac{di}{dt} = (-0.4) \times 5 \times (-3) \times e^{-3t}$$

$$V_0(t) = 6e^{-3t}\ V$$

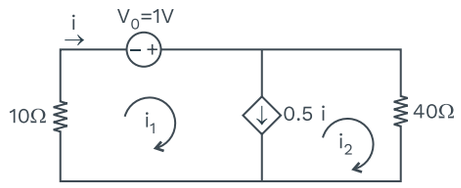
Example 13:

For the given circuit, obtain the value of current i(t) for time t > 0 if i(0) = 6A.



Solution:

Due to the presence of a dependent source, the voltage source V_0 = 1V is attached at the terminals of inductor as shown in the figure below.



Applying KVL in the outer loop, we get-

$10i_1 - 1 + 40\ i_2 = 0$

$10i_1 + 40\ i_2 = 1$ .... (1)

But, $i_1 - i_2 = 0.5\ i$ .... (2)

and $i_1 = i$ .... (3)

i.e. $i - i_2 = 0.5i$

$0.5i = i_2$

or $i_2 = 0.5\ i$ .... (4)

Value of i_2 and i_1 put in equation (1), we get-

$10i + 40\ (0.5\ i) = 1$

$10i + 20\ i = 1$

$$i = \frac{1}{30}$$

$$\therefore \quad R_{Th} = \frac{V_0}{i} = \frac{1}{1/30} = 30\,\Omega$$

The time constant is-

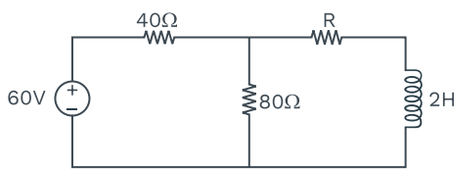
$$\tau = \frac{L}{R_{Th}} = \frac{6}{30} = \frac{1}{5} = 0.2\,\text{seconds}$$

$$i(t) = i(0)e^{-\frac{t}{\tau}}$$

or $i(t) = 6e^{-5t}A$

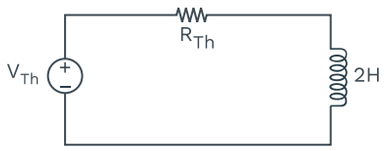
Example 14:

Calculate the value of resistance 'R' in the circuit for which the steady-state energy stored in the inductor is 1J.



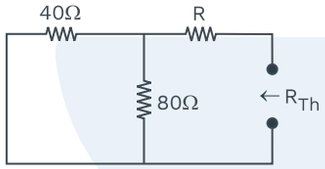
Solution:

The circuit can be replaced by its Thevenin equivalent is shown below.



$$V_{Th} = 60\left(\frac{80}{80+40}\right) = 40\,V$$

For R_{Th} :



$$R_{Th} = \frac{40\times 80}{40+80} + R = \frac{80}{3} + R$$

$$I = i(0) = i(\infty) = \frac{V_{Th}}{R_{Th}} = \frac{40}{(80/3)+R}$$

$$W = \frac{1}{2}Li^2 = \frac{1}{2}(2)\left(\frac{40}{\frac{80}{3}+R}\right)^2$$

$$\left(\frac{40}{80/3+R}\right)^2 = 1$$

$$\text{or } \frac{40}{80/3+R} = 1$$

$$\text{or } \frac{80}{3} + R = 40$$

$$R = 40 - \frac{80}{3} = \frac{40}{3} = 13.333\,\Omega$$

STEP RESPONSE OF A FIRST ORDER CIRCUIT

"When the dc source (either a voltage source or a current source) is applied to a circuit, it can be modeled as a step function, and the response is called as step response of the circuit."

Step response of a RC circuit [RC circuit with source]

- In a RC circuit when the DC source is applied suddenly, the voltage or the current source can be expressed as a step function. The response obtained is called as the step response.
- "The step response of a circuit is due to a sudden application of the voltage source or the current source".
- Consider a RC circuit as shown in the figure 5.12 (a), it can be replaced by the circuit in the figure 5.12 (b), where the voltage (V_S) is constant DC source.

So, $V(0^-) = V(0^+) = V_0$ .....(1)

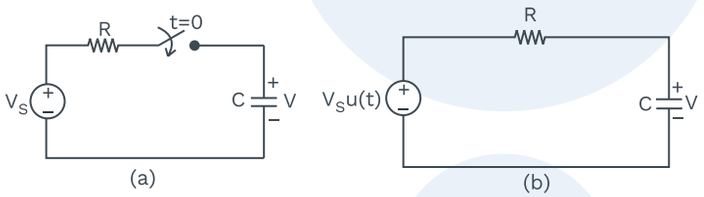


Figure 5.12: (a) RC circuit with switch (b) Equivalent RC circuit

Where the voltage $V(0^-)$ is the capacitor voltage before the switching and the voltage $V(0^+)$ is the immediate voltage after switching.
Applying KCL in the circuit, we get-

$$C\frac{dv}{dt} + \frac{v - V_S u(t)}{R} = 0$$

or $$\frac{dv}{dt} + \frac{v}{RC} = \frac{V_S}{RC}u(t) \quad \text{.....(2)}$$

Where, voltage (v) is the capacitor voltage.
For time t > 0, the switch is closed and equation (2) becomes,

$$\frac{dv}{dt} + \frac{v}{RC} = \frac{V_S}{RC}$$

$$\frac{dv}{dt} = \frac{1}{RC}[V_S - v]$$

or $$\frac{dv}{v - V_S} = \frac{-dt}{RC} \quad \text{.....(3)}$$

On integrating both the sides with the initial conditions, we get-

$$\int_{V_0}^{V(t)} \frac{dv}{v - V_S} = \int_0^t \frac{-dt}{RC}$$

$$\ln[v - V_S]\Big|_{V_0}^{V(t)} = \frac{-t}{RC}\Big|_0^t$$

$$\ln\left[\frac{(V(t) - V_S)}{(V_0 - V_S)}\right] = \frac{-t}{RC} + 0$$

$$V(t) - V_S = (V_0 - V_S)\, e^{-\frac{t}{RC}}$$

or $$V(t) = V_S + (V_0 - V_S)e^{-\frac{t}{RC}} \quad \text{.....(4)}$$

or $$V(t) = V_S + (V_0 - V_S)e^{-\frac{t}{\tau}}, t > 0 \quad \text{.....(5)}$$

where τ is time constant.
i.e. $\tau = RC$

This, $$V(t) = \begin{cases} V_0, & t < 0 \\ V_S + (V_0 - V_S)e^{-\frac{t}{\tau}}, & t > 0 \end{cases} \quad \text{.....(6)}$$

Equation (6) represents the complete response (or the total response) of a RC circuit when the DC voltage applied suddenly.
If the capacitor is charged initially, assumed that the source voltage is, $V_S > V_0$, as shown in the figure 5.13.

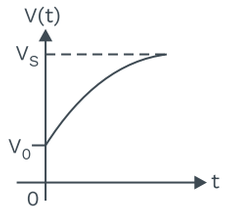


Figure 5.13: Response of RC circuit with initially charged capacitor

Definition

Step Response of Circuit:
"The nature of the circuit when the excitation is a step function, which can be a voltage or a current source is called as step response of the circuit".

Initially if the capacitor is not charged, then V_0 = 0V, so the equation becomes -

$$V(t) = \begin{cases} 0, & t < 0 \\ V_S\left(1 - e^{-\frac{t}{\tau}}\right), & t > 0 \end{cases} \quad \text{.....(7)}$$

The above equation can be expressed as -

$$V(t) = V_S\,(1 - e^{-t/\tau})\,u(t) \quad \text{.....(8)}$$

The current through capacitor is-

$$i(t) = C\frac{dV}{dt} = \frac{C}{\tau}V_S e^{-\frac{t}{\tau}},\ \tau = RC,\ t > 0$$

$$\text{or } i(t) = \frac{V_S}{R}e^{-\frac{t}{\tau}}u(t) \quad \text{.....(9)}$$

The capacitor voltage V(t) and current i(t) are represented in the plots shown in the figure 5.14 (a), and 5.14 (b) respectively.

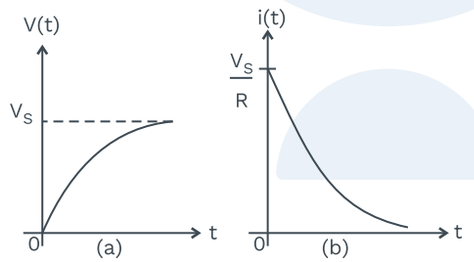


Figure 5.14: Step response of a RC circuit with an initially uncharged capacitor. (a) Voltage response (b) Current response

Complete Response:

Consider the equation (5),

$$V(t) = V_S + (V_0 - V_s)e^{-t/\tau}$$

This is complete response of RC circuit. It breaks into a natural response and a forced response.

$$V(t) = V_0 e^{-\frac{t}{\tau}} + V_S\left(1 - e^{-\frac{t}{\tau}}\right) \quad \text{.....(10)}$$

Complete Response = Natural Response (Stored energy) + Forced Response (Independent source)

or $V(t) = V_n(t) + V_f(t)$

Where,

$$V_n(t) = V_0 e^{-t/\tau}$$

and $$V_f(t) = V_S(1 - e^{-t/\tau})$$

The complete response can be obtained by the summation of the "transient response and the steady state response" i.e.

Complete Response = Transient response (Temporary part) + Steady State Response (Permanent part)

$$V(t) = (V_0 - V_S)e^{-t/\tau} + V_S \quad .....(11)$$

or $V(t) = V_t + V_{SS}$

Where, $V_t = (V_0 - V_S)e^{-t/\tau}$

and $V_{SS} = V_S$

- The transient response (V_t) or the temporary response is the part of complete response which decays to zero as the time approaches to infinite.
- The steady state response or the permanent response is the behavior of the circuit after the application of an external excitation.
- The complete response in equation (5) can be written in the form of the initial condition and final condition.

$$V(t) = V(\infty) + [V(0) - V(\infty)]e^{-t/\tau} \quad .....(12)$$

Where,

V(0) = Initial voltage at time t = 0.

V(∞) = Final or the steady state value.

If the position of the switch changes from time t = 0 to time t = t_0, then there is a time decay in the response.

So the equation (12) becomes-

$$V(t) = V(\infty) + \left[V(0) - V(\infty)\right]e^{-(t-t_0)/\tau} \quad .....(13)$$

Definition

Transient Response:
"The temporary response of a circuit that will die out with time is known as a transient response".

Definition

Steady State Response:
"The response of the circuit, a long time after the application of an external excitation is called as steady state response".

The key to working with RC circuit with source.

Step 1: To find the initial capacitor voltage V(0)

Step 2: To find the final capacitor voltage V(∞).

Step 3: To find the time constant τ.

Step 4: Substitute all the values in equation (13) and obtain the capacitor voltage V(t).

Step-1 is obtained for the circuit at time t < 0, and step-2 and step-3 are obtained from the circuit at time t > 0.

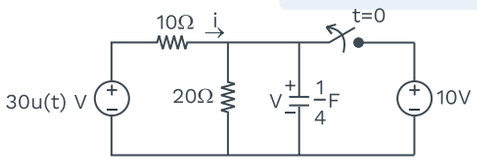
Keep in Mind:
Equation (12) or (13) applies only to step response u(t) or DC source.

Note:
If the switching is done at t = t_0, we will consider the equations-

$$V(t) = V(\infty) + \left[V(0) - V(\infty)\right] e^{-(t-t_0)/\tau}$$

Here, the initial conditions are determined at the instant t_0.

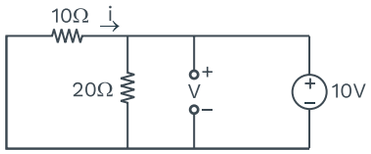
Example 15:
In the figure, the switch is closed for a long time is opened at t = 0. Find the value of current (i), and the voltage V for all time.



Solution:
The current 'i' can be discontinuous at time t = 0, while the voltage across the capacitor V can not.
From the unit step function,

$$30u(t) = \begin{cases} 0, & \text{for } t < 0 \\ 30, & \text{for } t > 0 \end{cases}$$

For time t < 0, the switch in the circuit is closed and 30u(t)=0V, i.e. 30u(t) voltage source is replaced by a short circuit. If the switch is closed for a long time, the capacitor behave as an open circuit at steady state condition. Thus, the circuit shown below in the figure (a) for time t < 0 is-



(a)

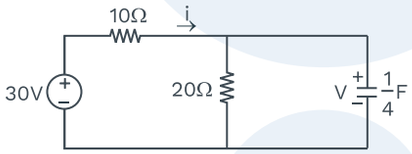
From the above circuit, we get-

$$V = 10 \text{ Volt}, \; i = \frac{-10}{10} = -1\,A$$

Since the voltage across the capacitor cannot be changed instantaneously, So,

$$V(0) = V(0^+) = 10V$$

For time t > 0, the switch will be opened and the voltage source will be removed from the circuit. The voltage source 30u(t) will be active in the circuit. So the circuit becomes as shown below in the figure (b).

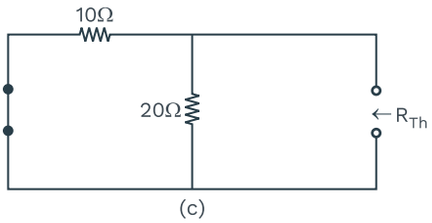


(b)

The circuit reaches to its steady state condition after a very long time and the capacitor in the circuit acts as an open circuit.
By using the voltage division rule, we get-

$$V(\infty) = 30 \times \frac{20}{20+10} = 20\,V$$

The Thevenin resistance at the capacitor terminal as shown in the figure (c) is-



(c)

$$R_{Th} = (10 \,||\, 20) = \frac{10 \times 20}{10+20} = \frac{20}{3}\,\Omega$$

and the time constant is-

$$\tau = R_{Th}C = \frac{20}{3} \times \frac{1}{4} = \frac{5}{3} \text{ seconds}$$

Thus,

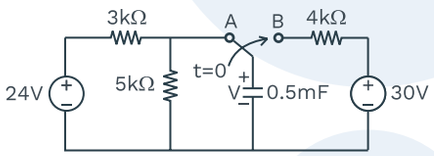
$$V(t) = V(\infty) + \left[V(0) - V(\infty)\right] e^{-t/\tau}$$

$$= 20 + \left[10 - 20\right] e^{-\frac{3}{5}t}$$

or $V(t) = 20 - 10e^{-0.6t}$ volts

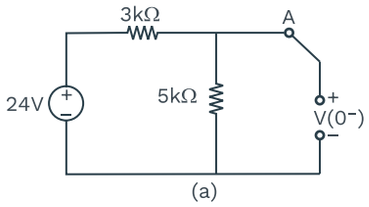
Example 16:

The switch present in the network has been in position 'A' for a very long time. At time t = 0, the switch moves from position 'A' to position 'B'. What is the value of voltage V(t) in the circuit for time t > 0 and also find the value of V(t) at time t = 1 seconds and t = 4 seconds.



Solution:

According to the question, the switch is in position 'A' for time t < 0, and so the capacitor in the circuit behaves as an open circuit as shown in below figure (a)-



(a)

Before time t = 0, the capacitor voltage is obtained by using the voltage division rule-

$$V(0^-) = \frac{5}{5+3}(24) = 15V$$

Since the capacitor's voltage cannot be changed instantaneously.

So,

$$V(0) = V(0^-) = V(0^+) = 15V$$

For time t > 0, the switch is at position B as shown in the figure (b).



(b)

Since, the Thevenin resistance is R_{Th} = 4 kΩ, so the time constant is-

$\tau = R_{Th}C$

$\tau = 4 \times 10^3 \times 0.5 \times 10^{-3}$

$\therefore \tau = 2$ seconds

At steady-state, capacitor behaves as an open circuit.

So, $V(\infty) = 30V$

$$V(t) = V(\infty) + \left[V(0) - V(\infty)\right]e^{-\frac{t}{\tau}}$$

$$= 30 + [15 - 30]e^{-0.5\,t}$$

or $V(t) = (30 - 15e^{-0.5t})$ Volts

At t = 1,

$V(1) = (30 - 15e^{-0.5})$

$V(1) = 20.9$ Volts

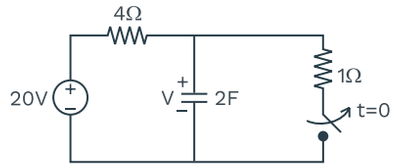
At t = 4,

$V(4) = (30 - 15e^{-2)}$

$V(4) = 27.97$ V

Example 17:

Calculate the capacitor voltage for t < 0 and t > 0 for the circuits in the figure.

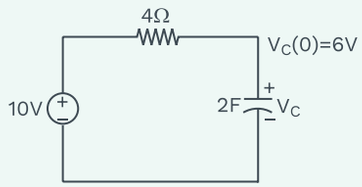


Solution:

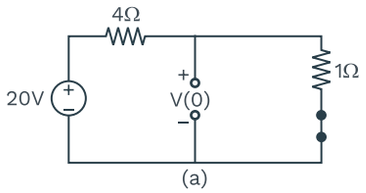
For t < 0, switch is closed. The capacitor acts like open circuit to DC as shown in the figure (a).

Previous Year's Questions

In the circuit shown in the figure, the energy absorbed by the 4Ω resistor in the time interval (0, ∞) is-



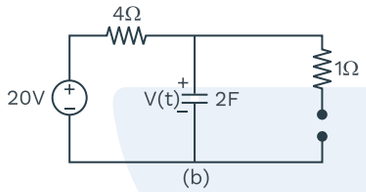
(1) 36 Joules
(2) 16 Joules
(3) 256 Joules
(4) None of the above



(a)

$$V(0) = \frac{20 \times 1}{4+1} = 4V$$

For t > 0, switch is open, as shown in the figure (b).



(b)

The Thevenin resistance connected to capacitor is R_{Th} = 4Ω and the time constant is-

$\tau = R_{Th} C = 4 \times 2 = 8$ seconds

At steady-state condition, the capacitor behaves as an open circuit.

So,

$V(\infty) = 20V$

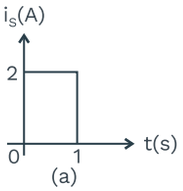
$V(t) = V(\infty) + [V(0) - V(\infty)]e^{-t/\tau}$

$V(t) = 20 + [4 - 20]e^{-t/8}$

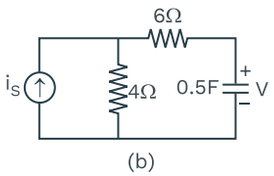
$V(t) = [20 - 16e^{-0.125t}]$ volts

Example 18:

If the waveform in the figure (a) is applied to the circuit in the figure (b). Find the voltage V(t) in the circuit. Assuming V(0) = 0 volt.



(a)



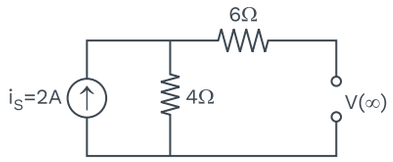
(b)

Solution:

Given, V(0) = 0

For 0 < t < 1,

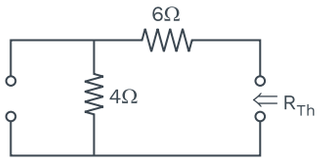
Since the capacitor acts like open circuit to DC at steady state as shown in the figure (a).



(a)

$V(\infty)=4\times2=8V$

The Thevenin resistance at capacitor terminal as shown in the figure (b)



(b)

$R_{Th}=4+6=10\Omega$

The time constant is-

$\tau = R_{Th}\, C = 10 \times 0.5 = 5$ seconds

$V(t) = V(\infty) + [V(0) - V(\infty)]e^{-t/\tau}$

$= 8 + (0 - 8)e^{-t/5}$

$= 8\,[1 - e^{-0.2t}]$ volts

For $t > 1$

$V(1) = 8[1 - e^{-0.2}] = 1.45$ V,

$V(\infty) = 0$ V,

Thus,

$V(t) = V(\infty) + [V(1) - V(\infty)]e^{-(t-1)/\tau}$

$= 0 + [1.45 - 0]e^{\frac{-(t-1)}{5}}$

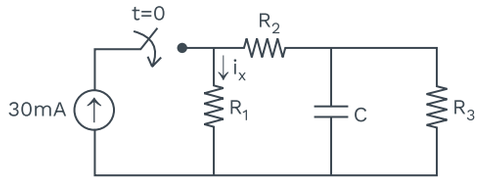
$V(t) = 1.45\, e^{-0.2(t-1)}$ volts

Thus,

$$V(t) = \begin{cases} 8\left(1 - e^{-\frac{t}{5}}\right) \text{V}, & 0 < t < 1 \\ 1.45e^{-\frac{(t-1)}{5}} \text{ V}, & t > 1 \end{cases}$$

Example 19:

For the network shown in the figure, determine the current i_x for time $t > 0$. Let the resistances, $R_1 = R_2 = 1k\Omega$, $R_3 = 2k\Omega$ and $C = 0.25$ mF.



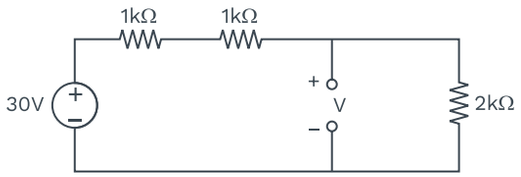
Solution:

For time t < 0, the switch is open.

Initial voltage of capacitor is-

$$V(0) = 0V$$

For time t > 0, the current source is transformed to the voltage source as shown below in the figure (a).



(a)

By using voltage division rule, we get-

$$V(\infty) = 30 \times \frac{2}{2+1+1} = 15V$$

$$R_{Th} = (1 + 1) \parallel 2$$
$$= 1\ k\Omega$$

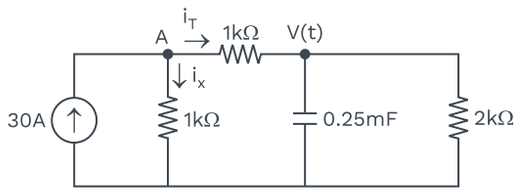
The time constant is-

$$\tau = R_{Th}C = 1 \times 10^{3} \times 10^{-3} \times 0.25 = 0.25 \text{ seconds.}$$

So,

$$V(t) = V(\infty) + [V(0) - V(\infty)]e^{-t/\tau}$$
$$V(t) = 15 + (0 - 15]e^{-4t}$$
$$V(t) = 15[1 - e^{-4t}] \text{ volts, } t > 0$$

We now obtain i_x from V(t), consider figure (b).



(b)

Applying nodal analysis at V(t).

$$-i_T + \frac{V(t)}{2k\Omega} + 0.25 \times 10^{-3} \frac{dV(t)}{dt} = 0$$

$$i_T = 7.5\times10^{-3}\left[1 - e^{-4t}\right] + 0.25\times10^{-3}\frac{d}{dt}\left[15\left(1 - e^{-4t}\right)\right]$$

$$= 7.5\times10^{-3}[1-e^{-4t}] + 3.75\times10^{-3}\times4e^{-4t}$$

$= 7.5[1-e^{-4t}] + 15e^{-4t}$ mA

or $i_T(t) = 7.5\ [1 + e^{-4t}]$ mA

Applying KCL at point (A), we get-

$30 = i_x + i_T$

or $i_x = [30 - i_T]$ mA

$i_x(t) = [30 - 7.5\ (1 + e^{-4t})]$ mA

$i_x(t) = [30 - 7.5 - 7.5e^{-4t}]$ mA

$i_x(t) = [22.5 - 7.5e^{-4t}]$ mA

$i_x(t) = 7.5\ [3 - e^{-4t}]$ mA, $t > 0$

Step response of an RL circuit [RL circuit with source]

"The response or the output of an RL circuit driven by the DC excitations is called as step response of the network".

- When the voltage source or the current source is applied suddenly in the RL network then the response of the circuit is termed as step response of an RL circuit.

The RL network shown in the figure 5.15 (a), is replaced by the network as shown in the figure 5.15 (b).

Thus, the current in an inductor can be obtained from the network shown below.

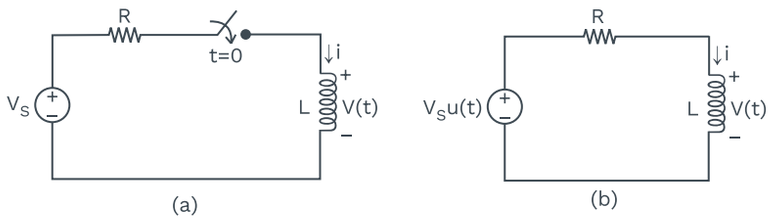


Figure 5.15: RL circuit with a step input voltage source

We know that the current through an inductor cannot be changed instantaneously. So,

$$i(0^+) = i(0^-) = I_0 \quad \text{....(1)}$$

where $i(0^-)$ is the current across the inductor just before switching and $i(0^+)$ is its current immediately after switching.

For $t > 0$, the switch is closed,

Applying KVL, we have-

$$-V_S u(t) + i(t)R + L\frac{di(t)}{dt} = 0$$

$$\frac{L}{R}\frac{di(t)}{dt} = \frac{V_S}{R}u(t) - i(t) \quad [u(t) = 1 \text{ for } t > 0]$$

$$\text{or } \frac{L}{R}\frac{di(t)}{dt} = \frac{V_S}{R} - i(t) \qquad \text{....(2)}$$

$$\text{or } \frac{di(t)}{\frac{V_S}{R} - i(t)} = \frac{R}{L}dt \qquad \text{....(3)}$$

On the integrating both sides with initial conditions, we get-

$$\int_{I_0}^{i(t)} \frac{di(t)}{\frac{V_S}{R} - i(t)} = \int_0^t \frac{R}{L}dt$$

$$-\ln\left[\frac{V_S}{R} - i(t)\right]\Big|_{I_0}^{i(t)} = \frac{R}{L}t\Big|_0^t$$

$$\text{or } \ln\left[\frac{V_S}{R} - i(t)\right] - \ln\left[\frac{V_S}{R} - I_0\right] = -\frac{Rt}{L}$$

$$\text{or } \ln\left[\frac{\frac{V_S}{R} - i(t)}{\frac{V_S}{R} - i_0}\right] = -\frac{Rt}{L}$$

$$\text{or } \frac{V_S}{R} - i(t) = \left(\frac{V_S}{R} - I_0\right)e^{-\frac{Rt}{L}}$$

$$\text{or } i(t) = \frac{V_S}{R} - \left(\frac{V_S}{R} - I_0\right)e^{-\frac{Rt}{L}}$$

$$\text{or } i(t) = \frac{V_S}{R} + \left(I_0 - \frac{V_S}{R}\right)e^{-\frac{Rt}{L}} \qquad \text{....(4)}$$

$$\text{or } i(t) = \frac{V_S}{R} + \left(I_0 - \frac{V_S}{R}\right)e^{-\frac{t}{\tau}} \qquad \text{....(5)}$$

Where,

τ is time constant-

i.e. $\tau = \frac{L}{R}$

Thus,

$$i(t) = \begin{cases} I_0, & t < 0 \\ \frac{V_S}{R} + \left(I_0 - \frac{V_S}{R}\right)e^{-t/\tau}, & t > 0 \end{cases} \quad \text{....(6)}$$

The above equation represents the complete response of an RL circuit as shown in the figure 5.16. So, the response in equation (5) can be written as-

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau} \quad \text{....(7)}$$

Where, i(0) is the initial value of current and i(∞) is the final value of current.

Note:

If the switching takes place in the circuit at time $t = t_0$ instead of time $t = 0$, then equation (7) becomes-

$$i(t) = i(\infty) + \left[i(0) - i(\infty)\right]e^{-(t-t_0)/\tau}$$

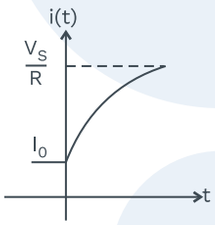


Figure 5.16: Total response of an RL circuit with an initial inductor current (I_0)

If the inductor is not charged initially, so the current in equation (6) will be $I_0 = 0$.

The current i(t) is-

$$\text{or } i(t) = \begin{cases} 0, & t < 0 \\ \frac{V_S}{R}\left(1 - e^{-t/\tau}\right), & t > 0 \end{cases} \quad \text{....(8)}$$

$$\text{or } i(t) = \frac{V_S}{R}\left(1 - e^{-t/\tau}\right)u(t) \quad \text{....(9)}$$

Equation (9) represents the step response of an RL circuit without any initial inductor current i_0. So, the voltage across an inductor is-

$$V(t) = L\frac{d}{dt}i(t)$$

$$V(t) = \frac{V_S L}{R\tau}e^{-t/\tau}, \qquad \tau = \frac{L}{R}, t > 0$$

$$\text{or } V(t) = V_S e^{-t/\tau}u(t) \quad \text{....(10)}$$

Figure 5.17 Shows the plots of inductor Voltage V(t) and Inductor current i(t).

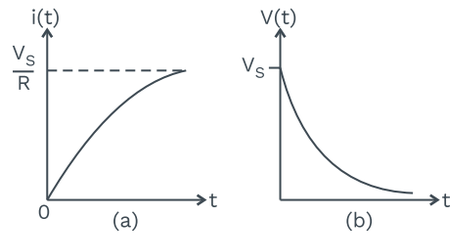


Figure 5.17: Step response of an RL circuit without initial inductor current
(a) Current response, (b) Voltage response

Complete Response

Consider the equation (5)-

$$i(t) = \frac{V_S}{R} + \left(I_0 - \frac{V_S}{R}\right)e^{-t/\tau}$$

This is complete response of RC circuit. It breaks into a natural response and a forced response.

$$i(t) = I_0 e^{-t/\tau} + \frac{V_S}{R}\left[1 - e^{-t/\tau}\right] \quad \text{....(11)}$$

Complete Response = Natural Response (Stored energy) + Forced Response (Independent source)

Definition

Natural Response:
"The behavior or response of current or the voltage in the circuit when all the external forces are set to zero".

or $i(t) = I_n(t) + i_f(t)$

where, $i_n(t) = I_0 e^{-t/\tau}$

and $i_f(t) = \frac{V_S}{R}\left(1 - e^{-t/\tau}\right)$

The complete response of a circuit is the summation of "Transient response and a steady state response i.e.".

Complete Response = Transient response (temporary part) + Steady State Response (Permanent part)

$$i(t) = \left(I_0 - \frac{V_S}{R}\right)e^{-t/\tau} + \frac{V_S}{R} \quad \text{....(12)}$$

or $i(t) = i_t + i_{SS}$ ....(13)

Where, $i_t = \left(I_0 - \frac{V_S}{R}\right)e^{-t/\tau}$

and $i_{SS} = \frac{V_S}{R}$

- The transient response or the temporary response of a circuit is a part of the complete response that decays to zero as the time approaches to infinite ($t > \infty$).
- The steady state response or the permanent response is the behavior of the circuit a long time after the application of external excitation.

Steps to solve an RL circuit with sources are-

Step 1: To find the initial inductor current i(0).

Step 2: To find the final inductor current i(∞).

Step 3: To find the time constant.

Step 4: Substitute all the values in equation (7) and obtain the inductor current i(t).

We obtain step-1 from given circuit for t<0, and step-2 and step-3 from the circuit for time t > 0.

Keep in Mind:

Stability of first order circuits:

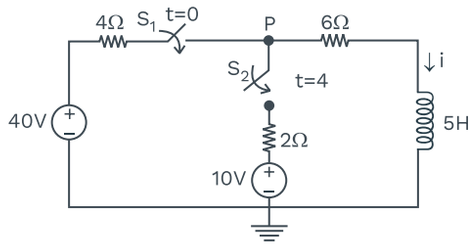
We know that $\tau = R_{Th}\ C$ or $\tau = \frac{L}{R_{Th}}$. So $\tau<0$ is possible when Thevenin equivalent resistance across capacitor or inductor is negative that is when $R_{Th} < 0$, the circuit is unstable.

Thus, for a first order circuit to stable.

$$R_{Th} > 0.$$

Example 20:

At time t = 0, switch-1 in the figure is closed, and switch-2 is closed 4 seconds later. Find current i(t) for time t > 0. Calculate the current i for 2s and t = 5s.



Solution:

Consider the three time intervals at $t \le 0$, $0 \le t \le 4$, $t \ge 4$ separately.

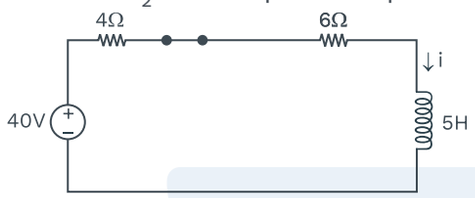
For time $t < 0$, the switch S_1 and S_2 in the circuit are open, so the current $i = 0$ A.

Since the current in an inductor cannot be changed suddenly.

So,

$$i(0^-) = i(0) = i(0^+) = 0$$

For $0 \le t \le 4$, S_1 is closed and S_2 is still open as represented in the figure (a).



(a)

The 4Ω and 6Ω resistors are in series, Since the inductor acts like short circuits to DC at steady state in the figure (a).

$$i(\infty) = \frac{40}{4+6} = 4\text{A}$$

The Thevenin resistance at inductor terminals,

$$R_{Th} = (4 + 6)$$

$$R_{Th} = 10\ \Omega$$

The time constant is-

$$\tau = \frac{L}{R_{Th}} = \frac{5}{10} = 0.5\,\text{seconds}$$

Thus,

$$i(t) = i(\infty) + \left[i(0) - i(\infty)\right]e^{-t/\tau}$$

$$i(t) = 4 + \left[0 - 4\right]e^{-2t}$$

$$\text{or } i(t) = 4\left[1 - e^{-2t}\right]\text{A}$$

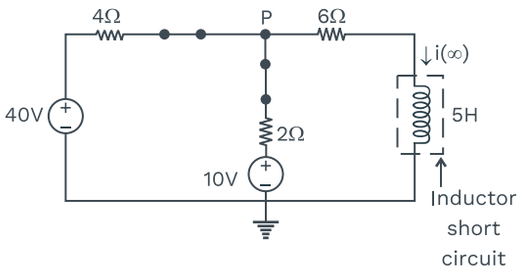
$$i(4) = 4\left[1 - e^{-(2\times4)}\right]\text{A}$$

$$\cong 4\text{A}$$

For time $t \ge 4$, switch S_2 is closed and the voltage source of 10V is connected to the circuit changes. The inductor current is not affected due to this sudden change because the current in an inductor cannot be changed suddenly.

So, the initial current will be-

$$i(4) = i\left(4^-\right) = i\left(4^+\right) \cong 4\text{A}$$



To find i(∞), let V be voltage at node P.
Applying KCL at the node P, we get-

$$\frac{V-40}{4}+\frac{V-10}{2}+\frac{V-0}{6}=0$$

$$\frac{3V+6V+2V}{12}=10+5$$

$$11V = 15 \times 12 = 180$$

$$V=\frac{180}{11}V$$

$$i(\infty)=\frac{V}{6}=\frac{180}{11\times 6}=\frac{30}{11}=2.727A$$

The value of Thevenin's resistance across the inductor terminals is-

$$R_{Th}=(4\parallel 2)+6=\frac{22}{3}\Omega$$

The time constant is-

$$\tau=\frac{L}{R_{Th}}=\frac{5}{\frac{22}{3}}=\frac{15}{22}\text{seconds}$$

Hence,

$$i(t)=i(\infty)+\left[i(4)-i(\infty)\right]e^{-(t-4)/\tau}$$

$$i(t)=2.727+\left[4-2.727\right]e^{-1.4667(t-4)}$$

$$i(t)=2.727+1.273\,e^{-1.4667(t-4)}\ A$$

$$i(t)=\begin{cases}0, & t\le 0\\ 4\left(1-e^{-2t}\right), & 0\le t\le 4\\ 2.727+1.273e^{-1.4667(t-4)}, & t\ge 4\end{cases}$$

Gray Matter Alert!!!

The zero initial conditions of a system means that the system is at rest and no energy stored in any of its elements or components.

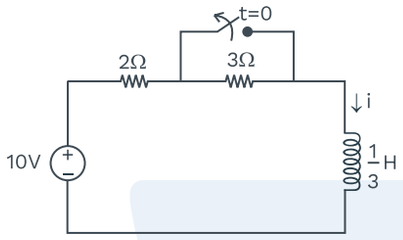
At time t = 2 seconds, and time t = 5 seconds

$i(2) = 4(1 - e^{-4}) = 3.93$ A

$i(5) = 2.727 + 1.273\, e^{-1.4667\times1} = 3.02$ A.

Example 21:

Obtain the value of current i(t) for the given network at time t > 0. Consider the switch is been closed for a very long time.



Solution:

For time t < 0, the 3Ω resistor will be short-circuited, and the inductor will behave as a short-circuit. The current through inductor at $t = 0^-$ (i.e. just before t = 0 is)-

$$i(0^-) = \frac{10}{2} = 5A$$

Since the current in an inductor cannot be changed instantaneously.

$i(0) = i(0^+) = i(0^-) = 5A$

For time t > 0, the switch is now open, the 2Ω resistor and a 3Ω resistor are in series. So,

$$i(\infty) = \frac{10}{2+3} = 2A$$

The equivalent resistance or the Thevenin's resistance across the terminal of an inductor is-

$R_{Th} = 2 + 3$

$R_{Th} = 5\Omega$

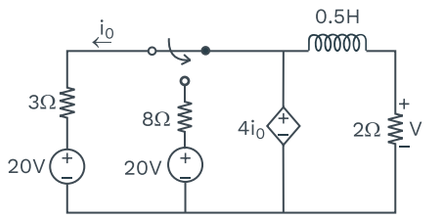
For the time constant,

$$\tau = \frac{L}{R_{Th}} = \frac{\frac{1}{3}}{5} = \frac{1}{15} \text{ seconds}$$

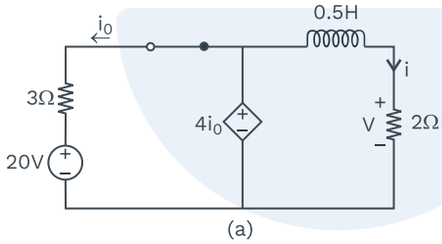
Thus, $i(t) = i(\infty) + \left[i(0) - i(\infty)\right]e^{-t/\tau}$

$i(t) = 2 + (5 - 2)e^{-15t} = 2 + 3e^{-15t}$A, t > 0

Example 22:
Find V(t) for t < 0, and t > 0 in the circuit of figure.



Solution:
For t < 0, consider the circuit shown in the figure (a).



(a)

Applying KVL in the circuit, we get-

$$-20 - 3i_0 + 4i_0 = 0$$

$$i_0 = 20$$

Since the inductors acts like to short circuits to DC at steady state in the figure (a).

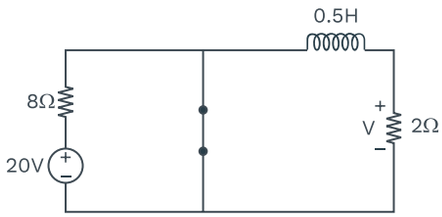
$$V(t) = 4i_0 = 80 = 20 \times 4 \text{ V}$$

$$i = \frac{V}{2} = \frac{80}{2} = 40 \text{ A}$$

For t > 0,
Consider the circuit in the figure (b).

$$i_0 = 0\text{A}$$

Hence, $4i_0 = 0$ then dependent voltage source short-circuited.



(b)

$i(\infty) = 0A$, $R_{Th} = 2\Omega$,

$$\tau = \frac{L}{R_{Th}} = \frac{0.5}{2} = 0.25\,\text{sec}$$

Thus,

$$i(t) = i(\infty) + \left[i(0) - i(\infty)\right]e^{-t/\tau}$$

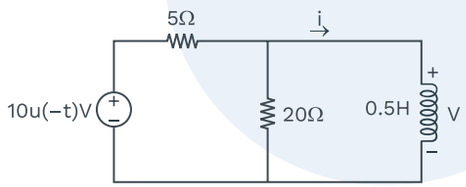
$i(t) = 0 + [40 - 0]e^{-t/0.25}$

$i(t) = 40e^{-4t}$ A

$V(t) = 2\ i(t) = 80e^{-4t}$ Volts

Example 23:

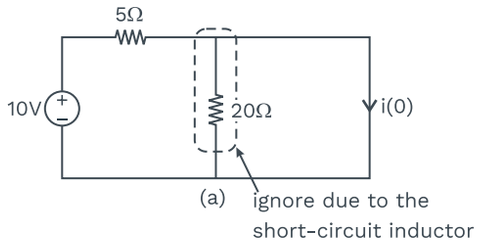
Obtain V(t) and i(t) in the circuit of figure.



Solution:

$$u(-t) = \begin{cases} 0, & t > 0 \\ 1, & t < 0 \end{cases}$$

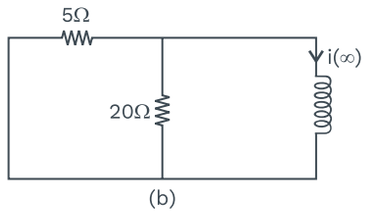
For t < 0, consider the circuit in the figure (a).



$$i(0) = \frac{10}{5} = 2\,A$$

$\therefore$ $i(0) = i(0^-) = i(0^+) = 2A$

for t > 0, consider the circuit in the figure (b).



(b)

$i(\infty) = 0A$

The Thevenin resistance at inductor terminals is-

$$R_{Th} = (5 \parallel 20) = \frac{5 \times 20}{20 + 5} = 4\Omega$$

Time constant $\tau = \frac{L}{R_{Th}} = \frac{0.5}{4} = \frac{1}{8}$ seconds

Thus,

$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}$

$i(t) = 0 + [2 - 0]e^{-8t}$

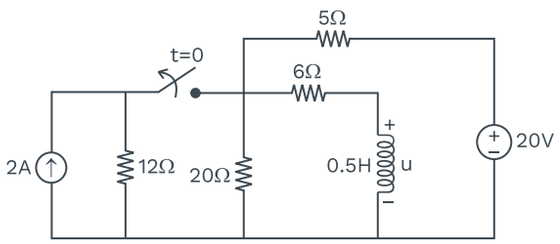
$i(t) = 2e^{-8t}$ A

$$V(t) = L\frac{di}{dt} = 0.5 \times 2 \times (-8)e^{-8t}$$

$V(t) = -8e^{-8t}\, u(t)$ Volts

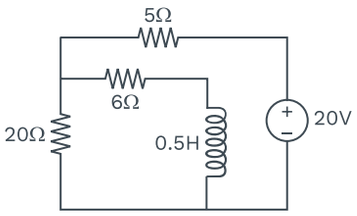
Example 24:

Find the time constant (τ) for the network shown in the figure.



Solution:

For t > 0, switch is open as shown in the figure.



The Thevenin resistance at inductor terminals is-

$$R_{Th} = (20 \| 5) + 6 = \frac{20 \times 5}{20 + 5} + 6 = 10\Omega$$

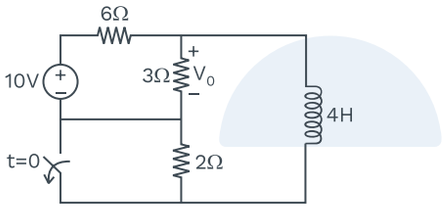
L = 0.5 H

The time constant is-

$$\tau = \frac{L}{R_{Th}} = \frac{0.5}{10} = 0.05 \text{ seconds}$$

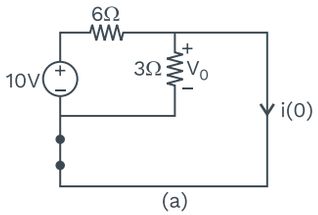
Example 25:

Find the voltage $V_0(t)$ for time t > 0 in the given network.



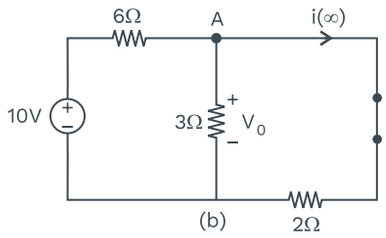
Solution:

For time t < 0, switch is closed consider the circuit in the figure (a). The inductor acts like short circuit and 2Ω resistor is ignored.



$$i(0) = \frac{10}{6} = 1.667 \text{ A}$$

For t > 0, switch open, consider the circuit in the figure (b).



Applying KCL at point A, we get-

$$\frac{V_0 - 10}{6} + \frac{V_0}{3} + \frac{V_0}{2} = 0$$

$$\frac{V_0 + 2V_0 + 3V_0}{6} = \frac{10}{6}$$

$$6V_0 = 10$$

$$V_0 = \frac{10}{6}$$

$$i(\infty) = \frac{V_0}{2} = \frac{10}{2 \times 6} = \frac{5}{6}$$

$R_{Th} = (3 \parallel 6) + 2 = 4\Omega$

The time constant (τ) is-

$$\tau = \frac{L}{R_{Th}} = \frac{4}{4} = 1 \text{seconds}$$

$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}$

$$i(t) = \frac{5}{6} + \left[\frac{10}{6} - \frac{5}{6}\right]e^{-t} = \frac{5}{6}\left[1 + e^{-t}\right] A$$

V_0 is the voltage across the 4H inductor and 2Ω resistor.

$$-V_0(t) + L\frac{di(t)}{dt} + 2i(t) = 0$$

$$V_0(t) = 2 \times \frac{5}{6}\left[1 + e^{-t}\right] - 4 \times \frac{5}{6}e^{-t}$$

$$= \frac{10}{6} + \frac{10}{6}e^{-t} - \frac{20}{6}e^{-t}$$

$$= \frac{10}{6}\left[1 - e^{-t}\right]$$

$V_0(t) = 1.6667\ [1-e^{-t}]$ Volts

Chapter Summary



- A circuit is 'first-order' because its behaviour is described by a first order differential equation.
- The capacitor behaves as an open-circuited to steady-state conditions while the inductor behave as short-circuited to the steady-state conditions.
- The natural response or the unforced response or the characteristic response is observed in a circuit when there is no external energy or no active power source present in the circuit.

$$x(t) = x(0)e^{-\frac{t}{\tau}}$$

- The time required for a response of a circuit to decay to $\frac{1}{e}$ or 36.7 % of the initial value is termed as the time constant (τ).
  For RC circuits, time constant τ = RC and for RL circuits, time constant $\tau = \frac{L}{R}$.
- The singularity functions such as unit step function, unit ramp function and unit impulse function.
  Unit step function u(t) is-

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$

  Unit impulse function $\delta(t)$ is-

$$\delta(t) = \begin{cases} 0, & t < 0 \\ \text{undefined}, & t = 0 \\ 0, & t > 0 \end{cases}$$

  Unit ramp function r(t) is-

$$r(t) = \begin{cases} 0, & t \le 0 \\ t, & t \ge 0 \end{cases}$$

- Complete response = Transient response (temporary response) + Steady state response (permanent response).
- The steady state response or the permanent response is the behavior of the circuit after the application of an independent source or an external excitation for a long time.

- The transient response or the temporary response is a part of complete response which delays to zero as time approaches to infinity.
- The total or complete response consist of steady state response and the transient response.
- The response obtained in the circuit due to the sudden application of DC source (either voltage or current) that can be modeled as a step function is called as step response. Finding the step response of a first order (in unit requires the initial value of $x(0^+)$ the final value of x (∞) and the time constant (τ). So, the obtained step response is-

$$x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}$$

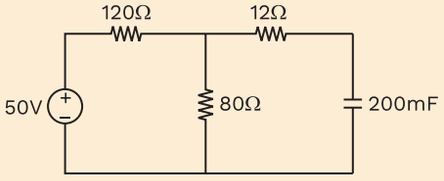
and the general form is-

$$x(t) = x(\infty) + \left[x\left(t_0^+\right) - x(\infty)\right]e^{\frac{-(t-t_0)}{\tau}}$$

Instantaneous value = Final + [Initial – Final] $e^{\frac{-(t-t_0)}{\tau}}$

PRACTICE QUESTIONS

Obtain the 'time constant (τ)' for the RC circuit is shown in the figure-



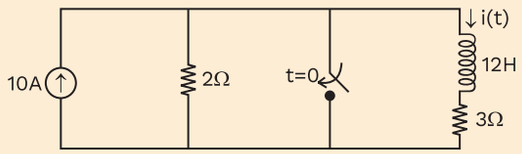
(A) 30 s **(B) 0.3 ms** **(C) 12 s** **(D) 300 ms**

Q2 **If 'voltage (V_S)' changes from 2V to 4V at t = 0, we can express (V_S) as-**

(A) 2u(t) V **(B) δ(t) V**

(C) [2u(-t) +4u(t)]V **(D) [2+2u(t)]V**

Q3 **In the circuit of figure, i(∞) is-**



(A) 0 A **(B) 10 A** **(C) 4 A** **(D) 6 A**

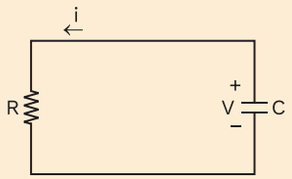
Q4 **Obtain 'time constant (τ)' for an RL circuit with resistance (R) = 10Ω and indunctance (L) = 20 H is-**

(A) 1 seconds **(B) 2 seconds** **(C) 0.5 seconds** **(D) 0.05 seconds**

Q5 **For the circuit shown below**

$V(t) = 56e^{-200t}$ V, t > 0

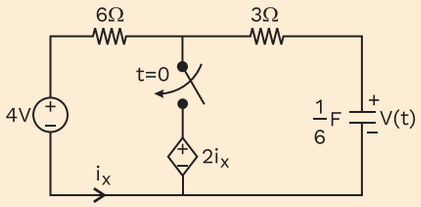
$i(t) = 8e^{-200t}$ V, t > 0



Obtain the time constant (τ).

(A) 200 sec **(B) 5 msec** **(C) 200 msec** **(D) 0.5 msec**

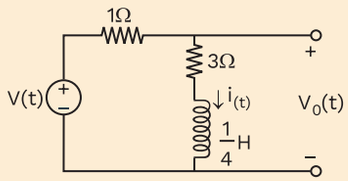
Q6 **For the shown in the figure, the switch is opened for a very long time and at time t = 0 the switch is closed, then the voltage V(t) obtained for time t > 0 is -**



(A) $(-2 + 2e^{-2t})$V **(B) $(2 - 6e^{-2t})$V**

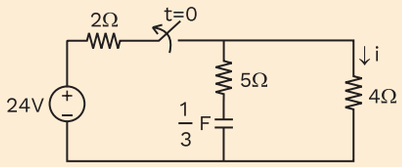
(C) $(2 + e^{-0.5t})$V **(D) $(-2 + 6e^{-2t})$V**

Q7 For the circuit shown in the figure, find $V_0(t)$, if i(0) = 2A and V(t) = 0.



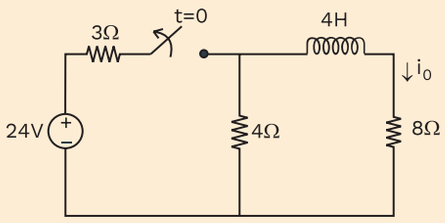
(A) $V_0(t) = -2e^{-16t}\,u(t)$ V
(B) $V_0(t) = 3e^{-0.12t}\,u(t)$ V
(C) $V_0(t) = 3e^{-t}\,u(t)$ V
(D) $V_0(t) = (e^{-t}+5)\,u(t)$ V

Q8 The RC circuit shown below. Calculate the current i(t), at t > 0.



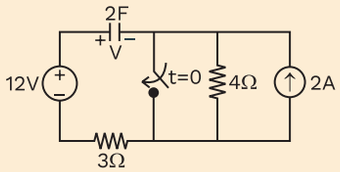
(A) $3e^{-t/3}$ A
(B) $1.5e^{-t}$ A
(C) $1.778e^{-t/3}$ A
(D) None of these

Q9 For the circuit in the figure, Find i_0 for t > 0.



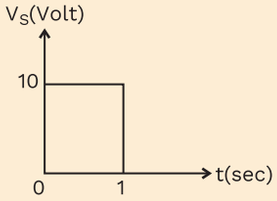
(A) 1.41 e^{-3t} A
(B) 5 $e^{-t/3}$ A
(C) 2 e^{-3t} A
(D) 5 e^{-3t} A

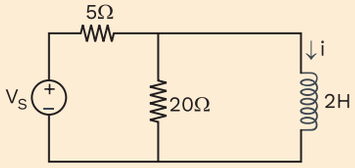
Q10 **Obtain the capacitor voltage (V_C) for t < 0 and t > 0 for the circuit shown in the figure-**



(A) $4V, (12 - 8e^{-t/6})\ V$
(B) $3V, -8e^{-t/6}\ V$
(C) $3V, (10 - 2e^{-t/6})\ V$
(D) None of these

Q11 **If the input pulse is applied to the circuit in the figure, determine the response i(t).**





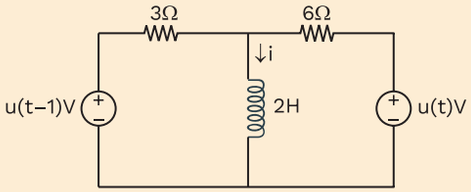
(A) $2\left(1-e^{-2t}\right)A, 0<t<1$
$1.729e^{-2(t-1)}A, t>1$

(B) $\left(3e^{-t}+5\right)A, 0<t<1$
$3A, t>1$

(C) $3e^{-t}A, 0<t<1$
$2e^{-2(t-1)}A, t>1$

(D) $5e^{-t}, 0<t<1$
$6e^{-t}A, t<1$

Q12 **Obtain the value of current i(t) in the network, if i(0) = 0 A.**



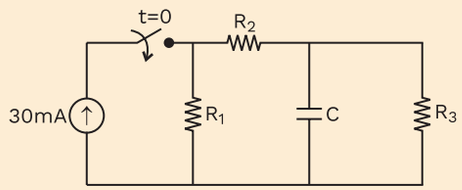
(A) $\frac{1}{6}\left(1-e^{-t}\right)A, 0<t<1$
$0.5-0.39e^{-(t-1)}A, t>1$

(B) $\left(5e^{-t}+1\right), 0<t<1$
$3e^{-t}, t>1$

(C) $3\left(1+3e^{-t}\right), 0<t<1$
$-39e^{-t}, t>1$

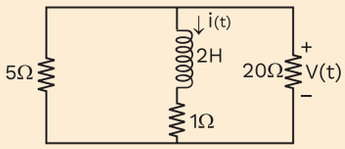
(D) $5e^{-t}+3, 0<t<1$
$\left(3e^{-t}+10\right), t>1$

Q13 **In the circuit shown in figure, find the time constant, if $R_1 = R_2 = 1k\Omega$, $R_3 = 2\,k\Omega$ and C = 0.25 mF.**



(A) 1 seconds **(B) 2 seconds**
(C) 3 seconds **(D) 0.5 seconds**

Q14 **Find i(t) for t > 0, in the circuit below, if i(0) = 10A.**



(A) $40\,e^{-2.5t}$ **A** **(B)** $10\,e^{-2.5t}$ **A** **(C)** $-40\,e^{-2.5t}$ **A** **(D)** $3\,e^{0.04t}$ **A**

ANSWER KEY

Q.1 (C)	**Q.2** (C, D)	**Q.3** (A)	**Q.4** (B)	**Q.5** (B)
Q.6 (D)	**Q.7** (A)	**Q.8** (C)	**Q.9** (A)	**Q.10** (A)
Q.11 (A)	**Q.12** (A)	**Q.13** (D)	**Q.14** (B)	